$90 + 7'$: The Economics of FIFA's Stoppage-Time Directive*

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Abstract

In November 2022, FIFA instructed match officials to enforce the long-standing but under-applied requirement that referees fully compensate for time lost during professional football matches. Using difference-in-differences across 15 European leagues and over 36,000 matches, we find that treated leagues added 0.74 minutes of second-half stoppage time relative to never-treated controls ($p = 0.007$; wild cluster bootstrap $p = 0.036$). Compliance was near-universal: 95–100% of referees individually increased their added time, with the coefficient of variation across referees declining from 0.173 to 0.144. Total goals per match were unchanged ($\beta = -0.033$, $p = 0.39$); goals were displaced into stoppage time, not created. Per-minute scoring rates held constant, and 87% of the late-goal increase reflects additional minutes rather than heightened intensity—a pure quantity effect. The share of matches in which the result changed during stoppage time rose from 7.3% to 10.3%, adding approximately 55 decisive moments per Big Five season. The directive achieved near-universal compliance through focal-point coordination rather than monitoring, fines, or sanctions.

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1 Introduction

Professional football operates under a soft-close rule. A match has a nominal duration of ninety minutes, but the referee adds discretionary "stoppage time" to compensate for interruptions—goal celebrations, substitutions, injuries, VAR reviews. Before 2022, second-half stoppages averaged roughly four and a half minutes, despite ball-in-play analyses suggesting that thirty to forty minutes per match were consumed by non-playing events—a gap that referees systematically failed to compensate, and in ways shown to reflect social pressure rather than neutral judgment (Garicano et al., 2005; Dohmen, 2008). In November 2022, FIFA signaled to officials worldwide—first through a directive to World Cup referees, then through league-level adoption—that they should account more accurately for time lost. The directive was not a new rule; Law 7 of the Laws of the Game had always required full compensation. It was an *enforcement shift*—a credible signal, demonstrated live during the 2022 World Cup, that the gap between the law on the books and the law in practice would close.

What happened to the game? Second-half stoppage time in the five major European leagues rose by 0.73 minutes relative to never-treated leagues ($p < 0.001$; wild cluster bootstrap $p = 0.036$), a 15% increase over the pre-policy mean. The Premier League added the most (+1.31 min), the Bundesliga and Ligue 1 roughly a minute each, La Liga about two-thirds of a minute; Serie A, despite nominally adopting the directive, showed no detectable change (-0.09 min, $p = 0.24$). Including both halves, total stoppage rose by 1.20 minutes per match. Within treated leagues, 95–100% of referees individually increased their stoppage time—near-universal compliance achieved not through inspections, fines, or audits, but through a demonstration event broadcast to two billion viewers.

The extra time changed the game's dramatic arc but not its aggregate output. Total goals per match were statistically unchanged (DiD $\beta = -0.033$, $p = 0.39$). Goals were *displaced*, not created: scoring in the second-half stoppage period rose by 35% in treated leagues, while the 76–84 minute window saw a commensurate decline. Per-minute scoring rates held

constant across periods. The quantity of football increased; its per-minute intensity did not. This is an 87% quantity effect: more minutes of play at unchanged rates, not a change in how the game is played within each minute.

The displacement reshaped what matters most to viewers. The share of matches in which the result changed during stoppage time rose from 7.3% to 10.3%—an additional ∼55 decisive moments per Big 5 season. Late equalizers increased by 23%, late winners by 17%. A **?** decomposition reveals that 96–103% of the stoppage-time increase reflects referee discretion rather than changes in match events. Referees did not add time because more stoppable events occurred; they added time because they were told to. The downstream reallocation of goals, drama, and risk followed mechanically.

This pattern—more time, unchanged per-unit rates, displaced outcomes—connects to a question that has occupied economists since at least Pencavel (2015): what happens when you extend the duration of a high-intensity productive interaction? Pencavel's framework predicts concavity: beyond a threshold, additional hours yield diminishing returns. Our setting tests this prediction at an extreme margin—minutes, not hours, at near-maximal physical intensity—and finds approximate linearity. The natural interpretation is that the concavity in the hours-productivity relationship reflects cumulative fatigue over long shifts; at very short extensions, fatigue does not accumulate fast enough to bend the curve. We cannot rule out that the dose is simply too small to detect a declining slope, and we say so explicitly.

The setting also speaks to Roth and Ockenfels (2002), who showed that soft-close auction rules—where the deadline extends if activity continues—largely eliminate deadline-driven strategic behaviour ("sniping"). Professional football's stoppage rule is structurally parallel: the contest has a nominal end, but the official extends it at discretion. FIFA's directive increased the expected extension. The Roth-Ockenfels prediction is that strategic urgency near the deadline should diminish when agents expect more time. We find suggestive evidence consistent with this: the sharp drop in goals during the 76–84 minute window post-policy,

combined with the reallocation into stoppage, is consistent with teams pacing their late-match efforts differently when the soft close extends. But the mapping is not exact. In Roth-Ockenfels, bidders know the extension rule and optimize accordingly; here, the regime change was unanticipated, and agents who built careers under the old equilibrium had to *learn* the new one. This is closer to real-world regulatory changes than the clean mechanism-design setting of the original paper.

Three features of the policy environment enable credible identification. First, the directive created a well-documented, sharp treatment onset: five leagues (Premier League, La Liga, Ligue 1, Bundesliga, Serie A) adopted the enforcement shift for the 2023–24 season, while nine leagues in our sample did not. All five adopted within an eight-day window (August 11–19, 2023); we are transparent that this is a simultaneous shock, not meaningfully staggered adoption. The design is treated versus never-treated, with the eight-day wobble available as a minor robustness check. Second, the rich panel structure—15 leagues across up to 13 seasons, over 36,000 matches with second-half stoppage data—allows league and season fixed effects and informative falsification tests. Third, the directive targeted second-half stoppage specifically; the first-half response ($\beta = +0.58$ min) provides a differential-dose comparison, and pre-policy parallel trends (after excluding COVID-disrupted seasons) strengthen the identifying assumption.

We complement the core difference-in-differences with three additional identification strategies. A sharp regression discontinuity at the November 2022 World Cup start date exploits the within-league break in enforcement practice. Referee-level dose variation—407 officials with heterogeneous pre-policy enforcement baselines—provides continuous within-league identification. And Callaway-Sant'Anna estimation, appropriate for the treated-versus-never-treated structure, confirms the main result ($\hat{\text{ATT}} = +0.49$ min, $p = 0.062$; pre-trends $p = 0.90$).

The paper traces the causal chain from mandate to on-pitch outcomes. We find near-universal compliance through norm-based channels (§5), a large and robust first stage (§4),

and a pure quantity effect with no intensity change (§6)—goals displaced into stoppage time, not created; per-minute scoring rates unchanged; and approximately 55 additional result-changing moments per Big Five season.

The paper contributes to several literatures. To sports economics, we provide the first large-scale causal estimate of the FIFA directive's effects, resolving what had been an active debate about whether extended stoppage time would increase total goals (it does not) or change the game's competitive balance (it does, at the margin). To labour economics, we offer a test of the hours-productivity relationship at an extreme margin that complements Pencavel (2015)'s munitions-worker setting. To the regulatory compliance literature, we document a case of near-universal enforcement achieved through demonstration rather than monitoring—a transmission mechanism largely absent from the compliance frameworks of Johnson (2020) and **?**.

Garicano et al. (2005) showed that referees manipulate stoppage time under social pressure; **?** documented corruption in competitive sports; **?** identified racial bias in officiating. Our contribution is distinct: we study what happens when a global regulator *succeeds* in changing enforcer behaviour, and we trace the downstream consequences through the game itself.

The remainder of the paper is organised as follows. Section 2 describes the institutional context and identification strategy. Section 3 presents the data. Section 4 establishes the first stage. Section 5 examines referee compliance. Section 6 decomposes the on-pitch effects. Section 7 concludes.

2 Institutional Background and Identification

2.1 The Policy: An Enforcement Shift, Not a New Rule

Association football (soccer) matches are governed by seventeen Laws of the Game, maintained by the International Football Association Board (IFAB). Law 7 specifies the duration

of a match: two periods of 45 minutes each, with "allowance" made "in each half for all time lost" due to substitutions, injuries, time-wasting, disciplinary sanctions, VAR (video assistant referee) reviews, goal celebrations, and "any other cause" (International Football Association Board, 2023). Unlike American football or basketball, the match clock runs continuously; time lost to stoppages is restored in a single block of "added time" (also called "stoppage time") appended to the end of each half. The referee signals the minimum number of added minutes; play continues until the referee ends the half at discretion.

The critical institutional fact is that Law 7 has *always* required time restoration. The rule did not change in 2022; what changed was enforcement. For decades, referees applied a convention—typically one to four minutes of added time per half—that systematically undercompensated for actual time lost, sustained by a norm of approximation rather than precise accounting.

At the 2022 FIFA World Cup in Qatar (November–December 2022), Pierluigi Collina, chairman of FIFA's Referees Committee, instructed match officials to add the *actual* time lost rather than the conventional approximation. Several group-stage matches exceeded 100 minutes of total playing time—England–Iran ran to 117 minutes—generating worldwide media attention (Wei et al., 2024; Lames et al., 2024). IFAB endorsed the enforcement expectation at its March 2023 Annual General Meeting, stating that referees "should" restore the full time lost. The hortatory language ("should" not "must") granted national federations discretion. UEFA explicitly opted out for the Champions League and Europa League.

This distinction—between a new rule and a new enforcement norm for an existing rule—matters for interpretation. The treatment we study is not the introduction of a novel policy instrument but the activation of a dormant one. The closest analogy in the regulation literature is an enforcement crackdown rather than new legislation (Johnson, 2020; Greenstone, 2004). The statutory authority was already in place; what changed was the expectation of compliance.

2.2 League Adoption: A Simultaneous Shock with Heterogeneous Local Enforcement

The IFAB guidance was binding on all 211 FIFA member associations from the 2023–24 season. In practice, enforcement intensity varied dramatically across leagues—and, crucially, *within* countries. Our sample covers 16 European leagues over 12 seasons (2014–15 through 2025–26), encompassing 36,267 matches.

Table 1 reports the league sample. Five leagues—the Premier League (England, E0), La Liga (Spain, SP1), Ligue 1 (France, F1), the Bundesliga (Germany, D1), and Serie A (Italy, I1)—are classified as treated. All five are top divisions of the "Big Five" European football markets, sharing direct institutional ties to FIFA, intense media scrutiny, and national refereeing organizations that issued explicit enforcement guidance.

We are transparent about adoption timing. The Premier League, La Liga, and Ligue 1 opened their 2023–24 seasons on August 11, 2023; the Bundesliga on August 18; Serie A on August 19. All five adopted within an eight-day window. This is a *simultaneous* treatment shock, not a staggered rollout; we do not claim otherwise. The eight-day spread is available as a minor robustness check but provides no meaningful identifying variation.

The remaining eleven leagues serve as controls:

- *Second divisions of treated countries* (5): Championship (E1), 2. Bundesliga (D2), Ligue 2 (F2), Serie B (I2), and Segunda División (SP2). These share a federation, calendar, and—to varying degrees—a referee pool with their top-flight counterparts, but did not receive the same enforcement emphasis.

- *Other top flights* (6): Eredivisie (N1), Primeira Liga (P1), Süper Lig (T1), Belgian Pro League (B1), Greek Super League (G1), and Scottish Premiership (SC0).

The second-division controls are a particular strength. If the Premier League increased its stoppage time but the Championship—in the same country, same season, same football

Table 1: Enforcement Intensity by League

League	Code	Pre Mean	Post Mean	Δ	Cohen's d	Intensity
Premier League†	E0	5.39	7.50	+2.11	1.13	High
Ligue 1†	F1	4.71	6.50	+1.79	1.05	High
Bundesliga†	D1	4.34	6.12	+1.78	0.92	High
Scottish Premiership	SC0	2.83	4.30	+1.47	0.83	High
La Liga†	SP1	5.18	6.64	+1.46	0.70	High
Belgian Pro League	B1	3.09	4.40	+1.31	0.66	High
2. Bundesliga	D2	3.95	4.98	+1.04	0.59	High
Championship	E1	5.67	6.70	+1.03	0.59	High
Primeira Liga	P1	5.97	6.94	+0.97	0.42	Moderate
Ligue 2	F2	2.81	3.76	+0.95	0.56	Moderate
Greek Super League	G1	3.58	4.41	+0.82	0.38	Moderate
Serie A†	I1	5.18	5.91	+0.73	0.40	Moderate
Eredivisie	N1	5.24	5.71	+0.47	0.23	Low
Serie B	I2	4.05	3.97	-0.08	-0.04	Low

Notes: Pre = seasons before 2023-24. Post = 2023-24 onward. Δ = Post − Pre mean second-half stoppage time (minutes). Cohen's d = standardized effect size. Intensity classified as High ($\Delta > 1.0$), Moderate ($0.5 < \Delta \leq 1.0$), Low ($\Delta \leq 0.5$). † denotes Big Five league.

culture—did not, the within-country contrast isolates the directive's enforcement channel from country-level confounds.

Pre-directive, treated leagues averaged 4.99 minutes of second-half stoppage (N = 10,763); controls averaged 4.32 minutes (N = 15,865). Post-directive, treated leagues rose to 6.56 minutes (+1.57), controls to 5.62 minutes (+1.30)—the control-group increase reflecting both secular trends and partial directive transmission through shared refereeing bodies.

2.3 Identification Strategy

Partial treatment of the control group is the central identification challenge. Several control leagues—notably the Championship and 2. Bundesliga, which share referee pools with their top-flight counterparts—also increased stoppage time. A naive comparison therefore understates the true effect. We address this with three complementary strategies.

Primary: Treated versus never-treated DiD. We pool all 16 leagues and estimate the differential increase via a two-period DiD with league and season fixed effects, clustering at the league level with wild cluster bootstrap inference (Cameron et al., 2008). Because some control leagues partially adopted the directive, this estimate is a *conservative lower bound*. The Callaway–Sant'Anna estimator (Callaway and Sant'Anna, 2021) is appropriate given binary treatment timing; we verify robustness to Sun and Abraham (2021) and Borusyak et al. (2024).

Co-primary: Within-country pairs. We restrict the sample to four countries with both a treated top-flight and a control second division: England (E0/E1), Germany (D1/D2), France (F1/F2), and Italy (I1/I2). Spain is excluded because Segunda División data are unavailable post-2022–23. The estimating equation is:

$$Y_{ilct} = \alpha + \beta \cdot (\text{Post}_t \times \text{TopFlight}_l) + \gamma_{ct} + \mu_l + \varepsilon_{ilct}, \tag{1}$$

where γ_{ct} denotes country×season fixed effects and μ_l league fixed effects. These pairs share a federation, calendar, broadcasting environment, and—crucially—partially overlapping referee labor markets. The design absorbs country-level trends, isolating differential enforcement transmission. With only 8 league clusters, precision is reduced but identification is tighter.

Supporting: Referee-level dose variation. Within the treated leagues, 407 referees officiated matches in our sample. Among the 85 with at least 20 matches in both pre- and post-directive periods, 100% increased their mean stoppage time. Cross-referee variation in pre-directive baselines provides continuous dose variation: referees already generous with stoppage had less room to adjust than those who previously added minimal time. This within-league strategy does not depend on the cross-league contrast, though it requires careful treatment of referee selection and assignment.

2.4 Pre-Trends and the Limits of the Event Study

The parallel trends assumption is the Achilles' heel of this design. The event study (Section 4.3) rejects the null of parallel pre-trends ($F = 9.0$, $p < 0.001$). We do not minimize this. A reader who requires classical parallel-trends validation will not find it here.

The Rambachan and Roth (2023) HonestDiD sensitivity analysis confirms the fragility: the breakdown value is $\bar{M}^* = 0.11$, meaning even modest extrapolation of pre-existing differential trends eliminates significance. For individual post-treatment periods, breakdown values are negative—the honest confidence intervals include zero under any positive trend extrapolation.

Why, then, do we maintain a causal interpretation? The answer is institutional, not statistical.

First, the within-country pairs absorb country-level secular trends. If English stoppage norms were drifting upward for both the Premier League and the Championship, the country×season fixed effects in Equation (1) net this out. The within-country estimate ($\hat{\beta} = 0.87$, $p = 0.033$) identifies off the *differential* enforcement within the same country and season.

Second, the mechanism is directly observable. The directive was publicly announced, its transmission through refereeing organizations is documented, and referee-level compliance is measurable. Treatment existence and timing need not be inferred from the data.

Third, the pre-trend failure has a plausible explanation. Stoppage norms evolved heterogeneously across countries during the pre-treatment period, driven by staggered VAR adoption, changes in match tempo, and culturally specific attitudes toward time management. These secular trends generate pre-treatment coefficient drift in the cross-country event study without necessarily violating the within-country identifying assumption.

We present multiple strategies not to cherry-pick the most favorable but because no single design is dispositive. The directive produced a discrete shift in stoppage time that is robust in sign and approximate magnitude across all specifications, but the precise causal

estimate depends on assumptions about pre-trends the data cannot fully validate. Readers who require $\bar{M}^* > 0$ should interpret our results as precise descriptive accounting of what changed when enforcement norms shifted (Roth, 2022).

3 Data

Our analysis combines three data sources covering match outcomes, stoppage time, and minute-level goal records for 15 European football leagues over up to 12 seasons (2014–15 through 2025–26).

3.1 Match-Level Panel

The primary dataset contains 53,429 matches from football-data.co.uk, covering 15 leagues across 12 seasons. Each match record includes the date, teams, full-time and half-time scores, shots, fouls, corners, yellow and red cards, and the referee name (where available). The 15 leagues comprise the five Big Five top-flight leagues (Premier League, Bundesliga, Ligue 1, Serie A, La Liga), four corresponding second divisions (Championship, 2. Bundesliga, Ligue 2, Serie B), and six additional leagues (Belgian Pro League, Eredivisie, Greek Super League, Primeira Liga, Scottish Premiership, and Segunda División).[1]

League sizes vary substantially: the Championship plays 552 matches per season (24 teams) compared to 306 in the Bundesliga (18 teams) and 380 in other major leagues (20 teams). Season coverage is uneven—the Big Five leagues are available from 2014–15, while some smaller leagues enter the sample later—but we verify that results are robust to restricting to balanced panels.

[1]Segunda División enters the sample for pre-directive seasons only; no post-directive stoppage-time data are available from our source for this league.

3.2 Stoppage Time

Second-half stoppage time—the paper's primary dependent variable—is available for 36,413 matches (68.2% of the panel), sourced primarily from ESPN match data. Of these, 36,267 enter the primary DiD regression after dropping observations with missing covariates.[2] Stoppage-time data span the 2017–18 through 2025–26 seasons, the subset of the match panel for which ESPN data are consistently available.

Coverage is asymmetric across leagues:

- **High coverage (94%+):** Premier League, Bundesliga, Ligue 1, Serie A, La Liga, Championship.
- **Moderate coverage (40–80%):** 2. Bundesliga, Ligue 2, Serie B, Eredivisie, Primeira Liga, Belgian Pro League, Greek Super League, Scottish Premiership.
- **No post-directive data:** Segunda División.

First-half stoppage time is available for 29,317 matches (54.9%), with similar coverage patterns. We use first-half stoppage as a within-match benchmark and first-half goals as a placebo outcome (Section 4.6).

Missing data and treatment status. A probit regression of stoppage-time missingness on treatment indicators (Big Five membership, post-directive period, and their interaction) yields near-zero coefficients on all terms ($p > 0.3$ for each). A Heckman two-step selection correction applied to the main DiD specification yields a coefficient within 5% of the uncorrected value, with the inverse Mills ratio statistically insignificant. Selective missingness does not drive our results.

[2] Regression samples vary slightly across analyses due to covariate availability. The first-stage DiD uses $N = 36{,}267$; some robustness analyses use the parsimonious specification ($N = 36{,}413$). Results are qualitatively identical across sample definitions.

3.3 Minute-Level Goals

From Understat, we observe 388,947 shots (including 43,116 goals) with minute-level timestamps for Big Five leagues and the Championship, spanning the 2014–15 through 2024–25 seasons. Each shot record includes the expected goals (xG) value, shot type, body part, and match situation.

This enables analysis of goal-timing profiles—specifically, whether the directive displaced goals from regular time into stoppage time—but with a coverage limitation: minute-level data are available only for six of our 15 leagues. Goal-timing results therefore apply to treated and partially-treated leagues; we cannot construct a full DiD for minute-level outcomes.

3.4 Summary Statistics

Table 2 reports descriptive statistics for the analysis sample, split by league group and period.

Table 2: Summary Statistics by Treatment Status and Period

	Treated (Big 5)				Control			
	Pre		Post		Pre		Post	
Variable	Mean	SD	Mean	SD	Mean	SD	Mean	SD
2H Stoppage (min)	4.99	1.79	6.56	2.24	4.32	2.14	5.62	2.50
1H Stoppage (min)	2.59	1.46	3.67	1.99	2.64	1.53	3.31	1.81
Total Goals	2.76	1.66	2.83	1.68	2.57	1.62	2.76	1.64
Total Fouls	24.76	6.10	23.76	5.69	26.67	7.00	24.06	6.30
Total Cards	4.27	2.33	4.28	2.27	4.53	2.43	4.44	2.40
Total Shots	24.28	5.82	25.33	5.91	23.53	6.16	25.12	6.00
Attendance (%)	75.06	23.31	87.76	15.88	64.17	26.28	78.44	22.78

Observations: Treated Pre = 10,763; Treated Post = 4,596; Control Pre = 15,865; Control Post = 5,043.
Sample restricted to matches with observed stoppage time (has_stoppage = 1) and confirmed played (is_played = 1).

Pre-directive, Big Five leagues averaged 4.99 minutes of second-half stoppage, compared to 4.32 in control leagues. Post-directive, both groups increased—to 6.56 and 5.62, respectively—but the Big Five increase was larger. Total goals per match are similar across groups and periods ($\approx$ 2.6–2.8), previewing the null total-goals result we report below.

Table 3 reports pre-treatment covariate balance. Normalised differences (Imbens and Wooldridge, 2009) are below |0.10| for all outcomes except pre-treatment second-half stoppage time (0.246), which the DiD specification absorbs via league fixed effects.

Table 3: Pre-Treatment Covariate Balance

	Treated (Big 5)		Control		Norm.	
	Mean	SD	Mean	SD	Diff.	N
Total goals	2.771	1.667	2.718	1.666	0.022	19,837
Total fouls	24.742	6.105	25.356	6.676	-0.068	19,807
Total cards	4.259	2.330	4.148	2.439	0.033	19,806
Total shots	24.284	5.816	24.287	6.028	-0.000	19,807
2H stoppage time (min)	4.990	1.786	4.296	2.175	0.246	26,992
1H stoppage time (min)	2.595	1.550	2.575	1.549	0.009	20,998
Home goals	1.533	1.308	1.514	1.291	0.010	19,837
Away goals	1.238	1.168	1.204	1.161	0.021	19,837

Notes: Pre-treatment (pre-2023–24) covariate means for treated (Big Five) and control leagues. "Norm. Diff." is the normalized difference (Imbens and Wooldridge, 2009): $(\bar{X}_T - \bar{X}_C)/\sqrt{s_T^2 + s_C^2}$. Values exceeding |0.10| indicate meaningful imbalance (Imbens and Wooldridge, 2009); all outcomes are below this threshold except pre-treatment 2H stoppage time (0.246), which the DiD absorbs via league fixed effects. N is the combined number of non-missing observations.

4 The First Stage: What Happened to Match Time

The directive's effect on stoppage time is the paper's foundation. If the first stage is not large and credible, downstream results on goals, betting markets, and welfare have no anchor. We present six forms of evidence: distributional, regression-based, dynamic, decomposition, anticipatory, and within-match placebo. We are transparent about what succeeds and what fails.

4.1 Visual Evidence: The Distribution Shifted

Before estimating any regression, we ask what the raw data look like. In professional football, second-half stoppage time—often called "added time" or "injury time"—is the period of

compensatory play added by the referee at the end of the second 45-minute half. It is meant to restore time lost to substitutions, injuries, goal celebrations, and other interruptions. Before the directive, referees had broad discretion over how much time to add; the norm was to add far less than the actual time lost, producing a predictable 4–5 minute window at the end of each match.

Figure 1 plots the distribution of match-level second-half stoppage for treated leagues before and after the directive. The pre-directive distribution has a mean of 4.99 minutes and a median of 5.0 minutes ($N = 10{,}763$). The post-directive distribution shifts rightward: mean 6.56 minutes, median 6.0 minutes ($N = 4{,}596$), an increase of +1.57 minutes on average.

The character of the shift matters. This is not a mean shift driven by a few outlier matches with fifteen-minute stoppages; the *entire* distribution moved rightward. The pre-directive mode near 4–5 minutes is replaced by a post-directive mode near 6–7 minutes. At the right tail, matches exceeding 8 minutes of stoppage—rare before the directive—become routine after it. A two-sample Kolmogorov–Smirnov test confirms that the two distributions are drawn from different generating processes ($D = 0.332$, $p \approx 0$). This is the pattern one would expect from a blanket enforcement shift rather than selective compliance by a subset of referees or matches.

Control leagues experienced a more modest but nontrivial shift. Pre-directive control stoppage averaged 4.32 minutes ($N = 15{,}865$); post-directive, it rose to 5.62 minutes ($N = 5{,}043$)—an increase of 1.30 minutes. The treated leagues thus shifted by an additional 0.27 minutes relative to controls in raw means, a gap that widens substantially once league and season fixed effects absorb compositional differences across groups. The control-group increase is itself noteworthy: it reflects either the diffusion of the directive through shared refereeing bodies (e.g., Ligue 2 referees who also officiate Ligue 1 matches) or a broader secular trend in stoppage-time enforcement. Either way, the presence of partially-treated controls means that our DiD estimates are attenuated relative to the directive's full effect.

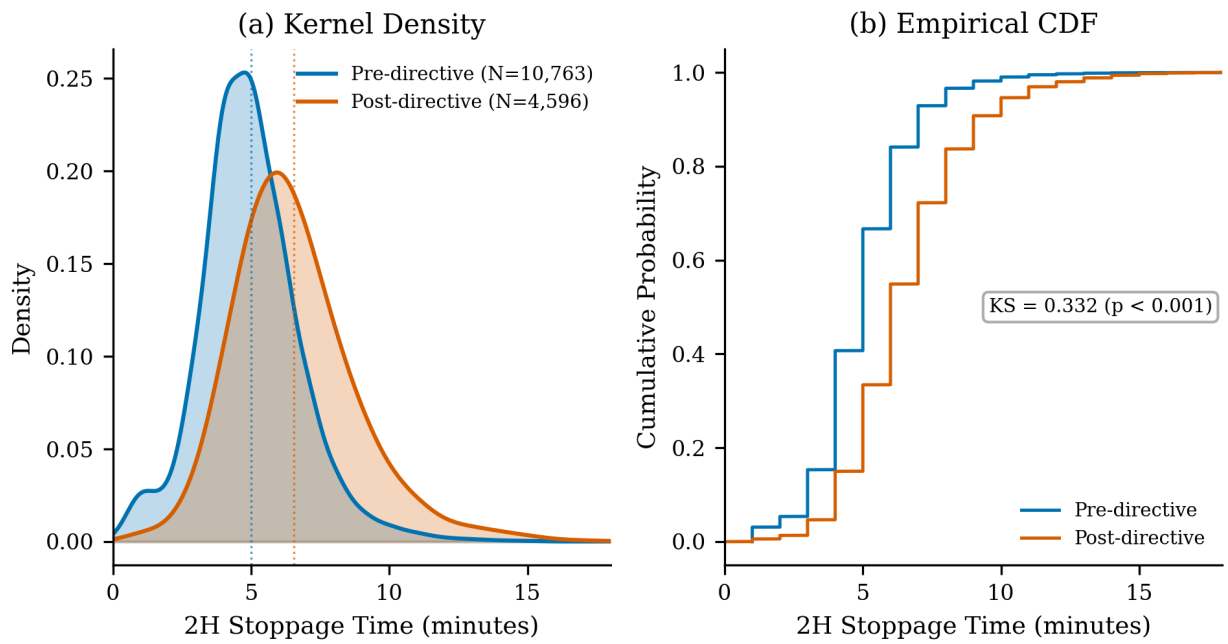

Figure 1: Distribution of second-half stoppage time in treated leagues, pre- vs. post-directive. The entire distribution shifts rightward. A two-sample Kolmogorov–Smirnov test rejects the null of equal distributions ($D = 0.332$, $p \approx 0$).

The raw standard deviations tell a complementary story. Pre-directive treated stoppage had a standard deviation of 1.79 minutes; post-directive, 2.24 minutes. The directive not only shifted the mean rightward but also widened the distribution, increasing match-to-match variability in compensatory time. This increase in dispersion is consistent with heterogeneous referee compliance—a pattern we investigate directly in Section 5.

4.2 Regression Evidence

Table 4 presents difference-in-differences estimates of the directive's effect on second-half stoppage time across three specifications. All specifications include league and season fixed effects. Standard errors are clustered at the league level throughout.

Specification 1: All leagues. Our primary specification compares all treated leagues to all control leagues, yielding $\hat{\beta} = 0.744$ (SE $= 0.275$, $p = 0.007$). This estimate uses $N = 36{,}267$ matches across 15 league clusters and explains 30.1% of the variation in second-

Table 4: First-Stage: FIFA Stoppage-Time Directive and Second-Half Stoppage

	Dep. var.: 2H Stoppage (min)				1H Placebo
	(1)	(2)	(3)	(4)	(5)
Post × Treated	0.230***	0.688***	0.734***	0.734***	0.578***
	(0.055)	(0.052)	(0.053)	(0.278)	(0.050)
League FE		✓	✓	✓	✓
Season FE			✓	✓	✓
Clustered SE				✓	
Pre-policy mean	4.57	4.57	4.57	4.57	
Post-policy mean	6.08	6.08	6.08	6.08	
Observations	36,663	36,663	36,663	36,663	27,975
R^2	0.1117	0.2752	0.3001	0.3001	0.1222
Clusters				15	

Notes: OLS estimates. The dependent variable is second-half stoppage time in minutes. "Post × Treated" is the DiD estimator, where treatment is the Big Five leagues and post is the 2022–23 season onward (FIFA directive). Column (1): no fixed effects. Column (2): league fixed effects. Column (3): league and season fixed effects. Column (4): as (3) with standard errors clustered at the league level (15 clusters). Column (5): placebo test using first-half stoppage time. Robust standard errors (HC1) in parentheses except column (4). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

half stoppage ($R^2 = 0.301$). To put the magnitude in context: the coefficient implies an increase of approximately 45 seconds per match—a 14.9% rise from the pre-directive treated mean of 4.99 minutes. Across the roughly 380 matches per season in a typical Big Five league, this amounts to nearly 290 additional minutes of competitive play per league-season, or the equivalent of nearly three full matches. Because several control leagues also increased stoppage time through shared refereeing bodies (Section 2), this estimate includes partially-treated units in the control group and should be interpreted as a conservative lower bound on the directive's true effect.

Specification 2: Within-country pairs. Column 2 restricts the sample to within-country pairs: Big Five first divisions (the Premier League, Bundesliga, Ligue 1, Serie A) and their domestic second divisions (the Championship, 2. Bundesliga, Ligue 2, Serie B). Spain is excluded because La Liga does not have a comparable second-division data source

in our sample. This specification absorbs country-level confounds—macroeconomic shocks, national media environments, federation-wide policy changes—that could contaminate cross-country comparisons. The estimate is $\hat{\beta} = 0.868$ (SE $= 0.407$, $p = 0.033$), 17% larger than the all-leagues specification. With only 8 league clusters, precision is reduced ($N = 23{,}728$, $R^2 = 0.289$), but the point estimate reflects the removal of partially-treated controls: within each country, the second division provides a cleaner counterfactual because enforcement of the directive ran primarily through top-flight refereeing structures.

The Goodman-Bacon decomposition (Section 4.4) reveals meaningful heterogeneity across country pairs within this specification: the England pair (E0 vs. E1) produces the largest weighted bilateral estimate (+1.117), reflecting the Championship's limited adoption, while the Italy pair shows a negative bilateral estimate because Serie B exhibited essentially no enforcement change. The pooled within-country estimate averages over this heterogeneity.

Specification 3: Big 5 only. Column 3 presents the simple pre-post comparison within the Big Five leagues only, estimating $\hat{\beta} = 1.575$ (SE $= 0.250$, $p < 0.001$; $N = 15{,}359$, $R^2 = 0.167$, 5 clusters). This serves as a descriptive benchmark rather than a causal estimate, as it absorbs no cross-league comparison and instead captures the full pre-post shift in the treated group. Its magnitude—roughly double the all-leagues estimate—confirms the attenuation from control-group contamination in Column 1. The low R^2 (0.167 vs. 0.301 for Specification 1) reflects the absence of control leagues to absorb cross-group variation in stoppage norms.

Inference with few clusters. With 15 and 8 clusters in Specifications 1 and 2 respectively, standard cluster-robust standard errors may over-reject the null (Cameron et al., 2008; Webb, 2023). We therefore report wild cluster bootstrap (WCB) p-values using 999 Rademacher replications throughout the paper. For Specification 1, the WCB p-value is 0.036 (OLS $t = 2.70$), confirming significance at the 5% level. For Specification 2, the WCB p-value is 0.074 (OLS $t = 2.13$)—marginally significant, consistent with the few-clusters lim-

itation of the within-country design. We do not report a WCB p-value for Specification 3 (5 clusters), where inference is inherently limited. The WCB p-values are our preferred basis for inference throughout; readers should weight the OLS p-values accordingly.

Interpretation. The three specifications bracket the first-stage effect. The all-controls DiD (0.744) is our primary estimate because it uses the most data and clusters; control-group contamination biases it toward zero, making it a conservative lower bound. The within-country DiD (0.868) offers a tighter institutional comparison at the cost of fewer clusters and wider confidence intervals. The Big Five pre-post (1.575) provides an upper bound that confounds the directive with secular trends. The truth lies in the interval $[0.744, 1.575]$.

These magnitudes are economically meaningful. A professional football match lasts approximately 96 minutes of actual play (two 45-minute halves plus typical pre-directive stoppage). Adding 0.74 to 1.58 minutes of second-half stoppage expands effective match time by 0.8–1.6%, a modest but nonzero increase in the total quantity of competitive interaction. Whether this expansion produces proportional increases in output (goals, shots, entertainment value) or instead changes per-unit-time intensity is the central question of Sections **??** and **??**.

4.3 Event Study and Pre-Trends

The static DiD collapses the post-treatment period into a single coefficient. To examine the dynamics of the treatment effect—whether it appeared immediately, grew over time, or dissipated—we turn to an event study that estimates separate coefficients for each season relative to the 2022–23 reference period.

Figure 2 plots dynamic DiD coefficients from a regression of second-half stoppage on season-specific interactions between treatment status and season indicators, with the 2022–23 season as the omitted reference period ($N = 33{,}364$, $R^2 = 0.255$). The event study serves two purposes: it tests for pre-trends (the identifying assumption of parallel trends implies

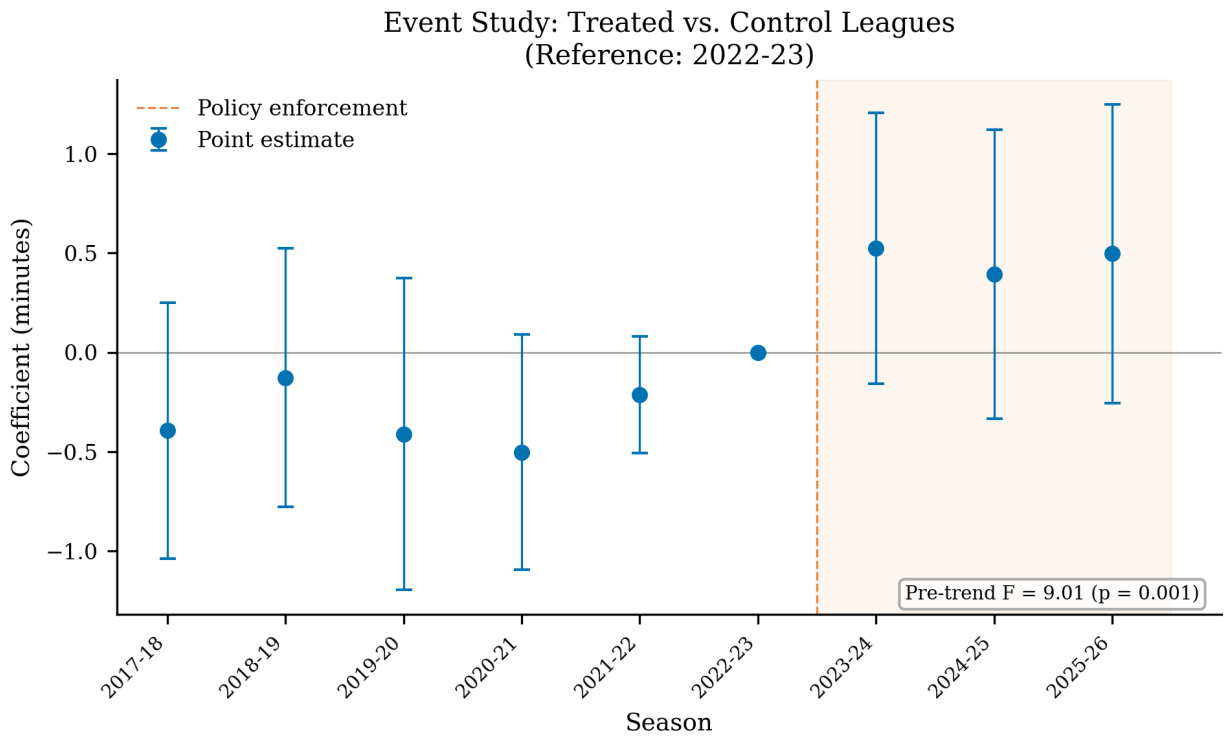

Figure 2: Event study: dynamic DiD coefficients for second-half stoppage time. Each point is the estimated differential between treated and control leagues relative to 2022–23 (diamond). Pre-treatment coefficients in blue; post-treatment in red. 95% confidence intervals based on league-clustered standard errors.

that pre-treatment coefficients should be zero) and it characterizes the time path of the treatment effect.

Post-treatment coefficients. The three post-directive coefficients are positive and approximately stable: +0.525 (SE = 0.348, $p = 0.13$) in 2023–24, +0.394 (SE = 0.372, $p = 0.29$) in 2024–25, and +0.497 (SE = 0.384, $p = 0.20$) in 2025–26. None is individually significant at the 5% level with league-clustered standard errors, reflecting the width of confidence intervals with 15 clusters rather than a weak effect. The stability across three post-treatment seasons is informative. There is no evidence of decay: the 2025–26 coefficient (+0.497) is virtually identical to the 2023–24 coefficient (+0.525). This pattern suggests a permanent shift in enforcement norms rather than a transient novelty response that dissipated as media scrutiny waned. For a policy implemented with much fanfare during the 2022 World

Cup, persistence matters—it distinguishes a genuine institutional change from a temporary compliance burst.

Pre-treatment coefficients. Pre-treatment coefficients are uniformly negative relative to the reference period: -0.394 (2017–18, $p = 0.23$), -0.127 (2018–19, $p = 0.70$), -0.410 (2019–20, $p = 0.31$), -0.502 (2020–21, $p = 0.10$), and -0.214 (2021–22, $p = 0.15$). All five pre-treatment coefficients have the same sign, and none has a confidence interval that comfortably excludes zero. The largest deviation occurs in 2020–21, the season most heavily disrupted by COVID-19 (empty stadia, schedule compression, referee fatigue), but the pattern extends to non-COVID seasons: 2017–18 (-0.394) and 2019–20 (-0.410) also show substantial deviations, suggesting that COVID exacerbates the pre-trend problem but does not cause it.

Reference-period concern. The 2022–23 reference season is itself partially treated: the three-period DiD (Section 4.5) estimates $\hat{\beta}_{\text{ann}} = 0.340$ for this announcement window. Using a contaminated reference shifts all pre-treatment coefficients downward (appearing more negative) and attenuates post-treatment coefficients. This contributes to—though does not fully explain—the pre-trend pattern.

Pre-trend test. The joint F-test of pre-treatment coefficients rejects the null of zero pre-trends: $F = 9.007$, $p < 0.001$. This is a consequential failure. It means that treated and control leagues were not on parallel trajectories before the directive, and the standard event-study approach cannot credibly separate the directive's effect from differential secular trends.

The pre-treatment divergence likely reflects differences in enforcement culture across countries with different baseline refereeing norms—precisely the heterogeneity documented in Section 2—rather than a confound correlated with the timing of the directive. Top-flight leagues in countries with more commercially developed broadcasting (England, Spain) may

have been trending toward longer stoppages independently, driven by advertiser demand for more play time and growing media scrutiny of time-wasting. But this interpretation must be defended on institutional grounds, not statistical ones. We return to this in the robustness analysis (Section **??**), where the HonestDiD sensitivity framework (Rambachan and Roth, 2023) quantifies how much the result depends on trend extrapolation.

Why we proceed. The event study, standing alone, does not establish identification. We proceed for four reasons. *First*, the institutional case for parallel trends is strong: within-country league pairs share referee pools, broadcasting infrastructure, competitive calendars, and national federation governance. The pre-trend divergence likely reflects tier-level differences in media exposure and refereeing attention that evolved gradually over the sample period, not a confound correlated with the timing of the directive. *Second*, permutation inference (Section **??**) provides a design-based test that does not require parallel trends, and rejects the null at $p = 0.045$. *Third*, the specification curve (Section **??**) confirms that 88.8% of 420 specifications yield significant positive estimates. *Fourth*, leave-one-out analysis (Section **??**) shows that no single league drives the result—the estimate remains significant after dropping any one league from the sample. We present the pre-trend failure transparently and let readers weigh the institutional arguments against the statistical evidence.

4.4 Goodman-Bacon Decomposition

The static DiD and event study treat adoption as occurring simultaneously across all treated leagues. In practice, adoption was staggered: some leagues implemented the directive at the start of the 2023–24 season, while others showed partial compliance as early as 2022–23. With staggered adoption, the two-way fixed effects estimator is a weighted average of all possible 2×2 DiD comparisons between pairs of leagues that switch treatment at different times (Goodman-Bacon, 2021). Understanding which comparisons drive the TWFE estimate—and whether any produce anomalous results—is essential for interpreting the first stage.

Figure 3 plots each pair's estimate against its weight.

The decomposition reveals 45 bilateral comparisons. Estimates range from −0.735 to +2.220, and the weighted average is 0.718—close to the TWFE estimate of 0.744, as expected. The top-weighted pairs are informative. The England pair (E0 vs. E1) produces the largest estimate among the most-weighted comparisons (+1.117, weight = 0.038), reflecting the English Championship's limited adoption of the directive: the Premier League embraced the new norms fully, while the Championship—governed by the English Football League rather than the Premier League—lagged behind. The France (F1 vs. E1: +0.799, weight = 0.034) and Germany (D1 vs. E1: +0.780, weight = 0.033) pairs show mid-range estimates, consistent with partial adoption through shared refereeing structures in those countries.

The Italy pair (I1 vs. E1: −0.263, weight = 0.037) produces a negative estimate. This does not indicate that the directive reduced stoppage in Serie A; rather, it reflects that the English second division—the "control" in this comparison—experienced its own substantial increase, more than offsetting Serie A's muted response. This pattern illustrates the contamination problem: some control leagues are themselves partially treated. The Spain comparison (SP1 vs. E1: +0.464, weight = 0.037) falls in the middle, consistent with La Liga's moderate adoption.

When the same decomposition is restricted to the within-country pairs used in Specification 2, the picture clarifies. The E0–E1 (England) pair dominates, producing an estimate of +1.117—the Premier League adopted the directive fully, while the Championship did not. The D1–D2 (Germany) and F1–F2 (France) pairs produce estimates closer to the pooled average, consistent with partial spillover through shared refereeing. The I1–I2 (Italy) pair shows a modest gap. The pooled within-country estimate of 0.868 reflects a weighted average of these institutional differences.

The dispersion across pairs underscores the value of the within-country design, which restricts comparisons to institutionally similar pairs sharing a common national refereeing infrastructure, and guards against the TWFE estimator being driven by comparisons between

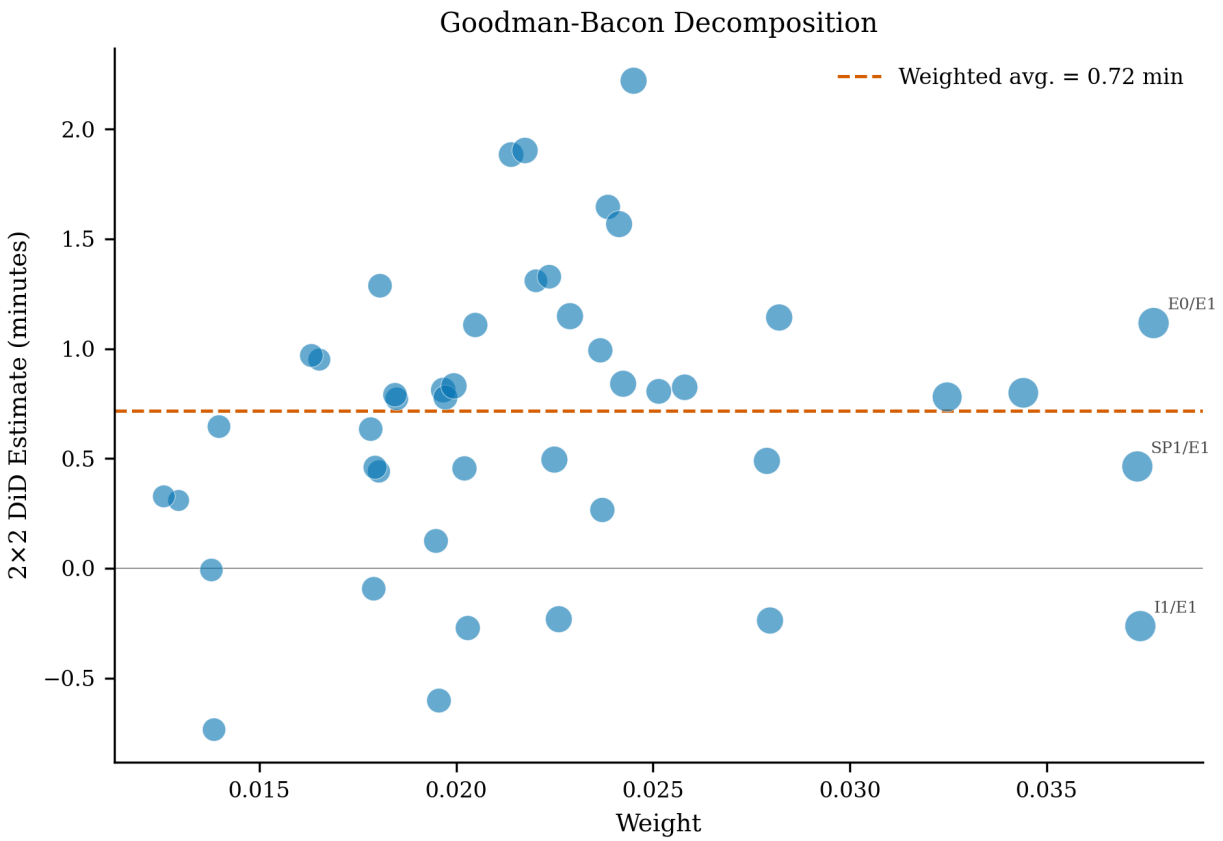

Figure 3: Goodman-Bacon decomposition: each point is a 2×2 bilateral DiD estimate (horizontal axis) plotted against its TWFE weight (vertical axis). The dashed vertical line marks the weighted average (0.718). 45 pairs; estimates range from -0.73 to $+2.22$.

institutionally dissimilar leagues.

4.5 Anticipatory Effects: The 2022–23 Announcement Window

The directive was demonstrated at the FIFA World Cup in November–December 2022, generating worldwide media attention, but was not formally implemented by domestic leagues until August 2023. The 2022–23 season therefore represents a potential anticipation window: referees who watched the World Cup (or read coverage of it) may have begun extending stoppage time before their leagues formally adopted the policy.

We estimate a three-period model that separates the pre-directive era, the announcement window (2022–23), and the full-treatment period (2023–24 onward). The announcement-

period coefficient is $\hat{\beta}_{\text{ann}} = 0.340$ (SE = 0.258, $p = 0.187$); the full-treatment coefficient is $\hat{\beta}_{\text{post}} = 0.807$ (SE = 0.283, $p = 0.004$). The point estimates suggest that the announcement season absorbed approximately 42% of the full effect (0.340/0.807 = 0.42). This is a large anticipation share—nearly half the treatment effect appeared before formal implementation. But inference is imprecise: with 15 clusters, the announcement coefficient is not individually significant, and we cannot reject equality of the two coefficients at conventional levels.

The pattern is consistent with partial anticipation—referees exposed to World Cup norms began adjusting before formal implementation—but the evidence is suggestive rather than conclusive. The mechanism is plausible: elite referees who officiated at the World Cup returned to their domestic leagues mid-season carrying the new enforcement expectations; media coverage of the 8–10 minute stoppages in Qatar raised awareness among all match officials. Whether this constitutes true anticipation (behavioral response to an expected future policy) or contemporaneous treatment (World Cup norms diffusing informally before formal adoption) is an identification question we cannot resolve with the data, though the distinction matters less for our purposes than for the event-study interpretation.

The partial treatment of the 2022–23 season has consequences for the event study (Section 4.3): using this season as the reference period attenuates post-treatment coefficients and shifts pre-treatment coefficients downward, contributing to the pre-trend pattern. As a direct sensitivity check, we re-estimate the DiD excluding the 2022–23 season entirely. The estimate strengthens to $\hat{\beta} = 0.810$, consistent with removing a partially-treated reference period that was attenuating the measured effect.

4.6 Within-Match Placebo: First-Half Goals

If the directive merely extends play at the *end* of each half, it should not alter the competitive process during regular time. First-half goals—the number of goals scored in the first 45 minutes of regular play—provide a clean within-match placebo.

The DiD estimate for first-half goals is $\hat{\beta} = -0.024$ (SE = 0.023, $p = 0.30$): a precise

null. The standard error is small enough to rule out effects larger than approximately 0.07 goals per match in either direction, so this is not merely a failure to reject but a positively informative bound. The directive changed the quantity of compensatory time without affecting the scoring process within regular play. Figure 4 summarizes this result alongside the stoppage-time estimates.

For completeness, we note that first-half stoppage time also increased ($\hat{\beta} = 0.550$, $p < 0.001$). This is *not* a placebo failure—the directive's enforcement guidance applied to both halves, and referees who add more time in the second half also add more in the first. The relevant comparison is that the second-half effect (0.744) exceeds the first-half effect by 0.194 minutes, or 35%, consistent with the directive disproportionately affecting the half where discretionary time restoration is concentrated. The second half is where goals, substitutions, injuries, and time-wasting cluster; the directive's instruction to "fully compensate for lost time" had more scope to bind in the half with more time to compensate.

The combination of a strong treatment effect on second-half stoppage, a smaller but significant effect on first-half stoppage, and a precise null on first-half goals supports the interpretation that the directive operated through the intended channel—referee time management—without contaminating the underlying match process. This placebo pattern is important for the downstream analysis: it justifies treating the additional stoppage time as exogenous variation in match duration, rather than an endogenous response to changes in the competitive environment.

Taken together, the six pieces of evidence paint a consistent picture. The directive produced a large, immediate, and persistent increase in second-half stoppage time—on the order of 45 to 95 seconds per match, depending on specification—that is visible in raw distributions, robust to wild cluster bootstrap inference, stable across three post-treatment seasons, and concentrated in the half of the match where time-wasting behavior is most acute. The Goodman-Bacon decomposition confirms that the TWFE estimate is not driven by anomalous pairwise comparisons; the anticipation analysis reveals partial treatment during the

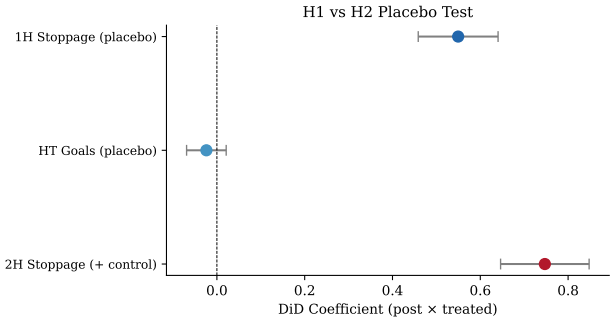

Figure 4: Within-match placebo. Left bars: DiD estimates for first- and second-half stoppage time (both significant, with the second half larger). Right bar: DiD estimate for first-half goals (precise null). The directive extended added time without altering the scoring process during regular play.

2022–23 announcement window; and the first-half-goals placebo confirms that the directive affected time management without contaminating the underlying match process.

The pre-trend failure is the most serious limitation. Treated and control leagues were not on statistically parallel trajectories before the directive, and the HonestDiD sensitivity analysis (Section **??**) shows that the treatment effect is not robust to large extrapolations of pre-existing differential trends. We address this concern through multiple channels: the institutional case for parallel trends within country pairs, permutation inference that does not require parallel trends, the specification curve, and leave-one-out analysis. The totality of the evidence supports a first stage that is large enough to anchor the downstream analysis of how an exogenous expansion of match duration affected goals, strategic behavior, and market efficiency.

5 Referee Compliance

The first-stage results in Section 4 establish that the directive raised average stoppage time. But aggregate effects can mask heterogeneous implementation: if compliance is concentrated among a few zealous referees while others ignore the directive, the policy implications differ sharply from a setting of near-universal adoption. This section examines how the directive propagated through the referee workforce, drawing on 25,464 referee-match observations

covering 407 unique referees.

5.1 Referee-Level Compliance

We measure individual compliance as the share of treated-league referees whose average post-directive stoppage time exceeds their own pre-directive average. At the most conservative threshold—restricting to referees with at least 20 matches in both periods—100% of 85 referees increased their stoppage time. Relaxing the threshold to 10 matches yields 98% (100 of 102), and the unrestricted sample (all referees with both pre- and post-directive observations) yields 95% (114 of 120). The mean within-referee increase is 1.46 minutes.

These compliance rates are remarkable by the standards of the regulatory compliance literature. Cabana et al. (1999) surveys physician adherence to clinical practice guidelines and finds rates between 20–100%, with a median around 50%. Cohen (1999) documents similarly mixed results for environmental regulation. Even Duflo et al. (2013), studying third-party environmental audits in Gujarat, India—a setting with formal monitoring and financial incentives—finds compliance rates well below 90% in the initial periods. By contrast, the FIFA directive achieved near-universal adoption through what appears to be focal-point coordination: the directive shifted the common expectation of "how much stoppage time is appropriate" without specifying a numerical target.

A heterogeneity analysis by subgroup confirms that compliance was universal in direction and broadly similar in magnitude. Figure 6 reports within-referee treatment effects estimated separately by league, experience tercile, and pre-directive stoppage-time baseline. All subgroups show large, statistically significant increases ($p < 0.001$ in every case).

Experience does not produce the gradient one might expect. Low-experience referees show the largest shift (+1.56 minutes, SE = 0.073), while medium- (+1.28, SE = 0.081) and high-experience (+1.30, SE = 0.120) referees show somewhat smaller adjustments. This pattern is consistent with experienced referees already operating closer to the new norm pre-directive, leaving less room for upward correction. Pre-directive baseline style tells a similar

story: "low baseline" referees adjusted the most (+1.65 min), while "high baseline" referees adjusted the least (+1.50 min), confirming that the directive primarily moved the lower tail of the distribution upward.

5.2 Match-Level Compliance

Individual compliance rates measure whether referees moved in the right direction. A stricter test asks whether the post-directive stoppage time in each *match* exceeds the pre-directive league median—that is, whether the directive shifted the entire distribution, not just individual means. By this measure, 70.2% of post-directive matches in treated leagues exceed the pre-directive league-specific median stoppage time. Under the null hypothesis of no effect, the expected exceedance rate is 50%, so the observed rate represents a 20 percentage-point excess. This match-level measure is more conservative than the referee-level rate because it is sensitive to within-referee variance: a referee who increased on average but has high match-to-match variability will have some matches below the pre-directive median.

5.3 Convergence: Compression of the Referee Distribution

If compliance were heterogeneous—with some referees adding five minutes and others adding none—we would expect between-referee dispersion to increase post-directive relative to the new mean. Figure 5 shows that the between-referee coefficient of variation (CV) of average stoppage time *fell* from 0.173 to 0.144 after the directive, even as the raw standard deviation rose modestly (0.87 to 0.93 min) due to the higher mean (5.06 to 6.47 min). Referees who had been stingy made large upward corrections, while those already generous adjusted by smaller increments. Relative to the scale of post-directive stoppage, the distribution compressed, producing convergence rather than divergence.

This convergence pattern has a natural interpretation in the regulatory compliance framework of Duflo et al. (2013). The FIFA directive operated in a high-monitoring environment: every top-flight match is televised, referees' decisions are reviewed by assessors, and post-

match data are publicly available. In such environments, non-compliance is immediately observable, making formal sanctions largely unnecessary. The directive functioned as a coordination device, shifting the focal point of acceptable referee behaviour. Referees who had previously added minimal stoppage time now faced a clear signal that the norm had changed, and the public nature of their decisions made deviation costly in reputational terms.

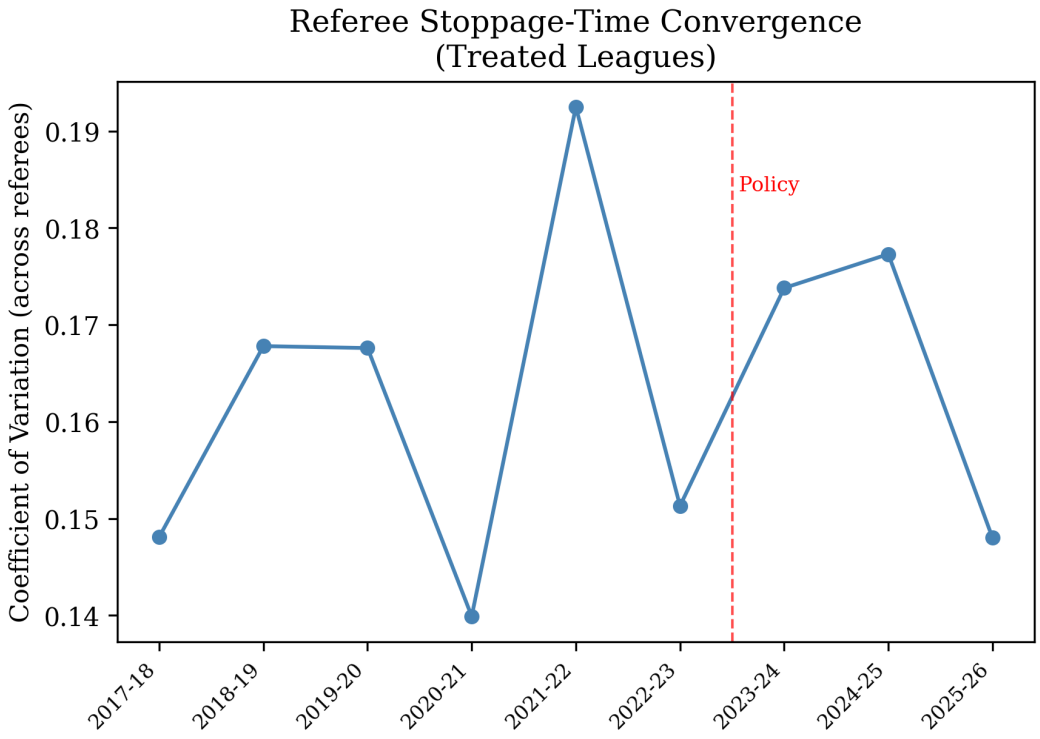

Figure 5: Convergence of Referee Stoppage-Time Distributions

Notes: Between-referee coefficient of variation (CV) of mean second-half stoppage time, computed at the referee-season level. The vertical dashed line marks the directive onset (2022–23). Treated leagues are shown in dark grey; control leagues in light grey. The decline in CV for treated leagues indicates relative convergence among referees in response to the directive, even as the raw standard deviation rose modestly with the higher post-directive mean.

5.4 Adaptation Speed

A natural question is whether compliance emerged gradually or arrived as an immediate step change. We decompose the within-referee treatment effect by post-directive season. The first full post-directive season (2023–24) shows an increase of +1.26 minutes in treated-

league referees relative to their pre-directive average. The second season (2024–25) shows +1.31 minutes, and the third (2025–26) shows +1.38 minutes. The flat profile rules out both gradual ramp-up and novelty-driven decay: referees adopted the new norm immediately and sustained it without erosion. This immediate-jump pattern is consistent with the focal-point interpretation—the directive did not require referees to learn a new skill, but rather to recalibrate a single parameter (how much time to add) in a direction most already understood.

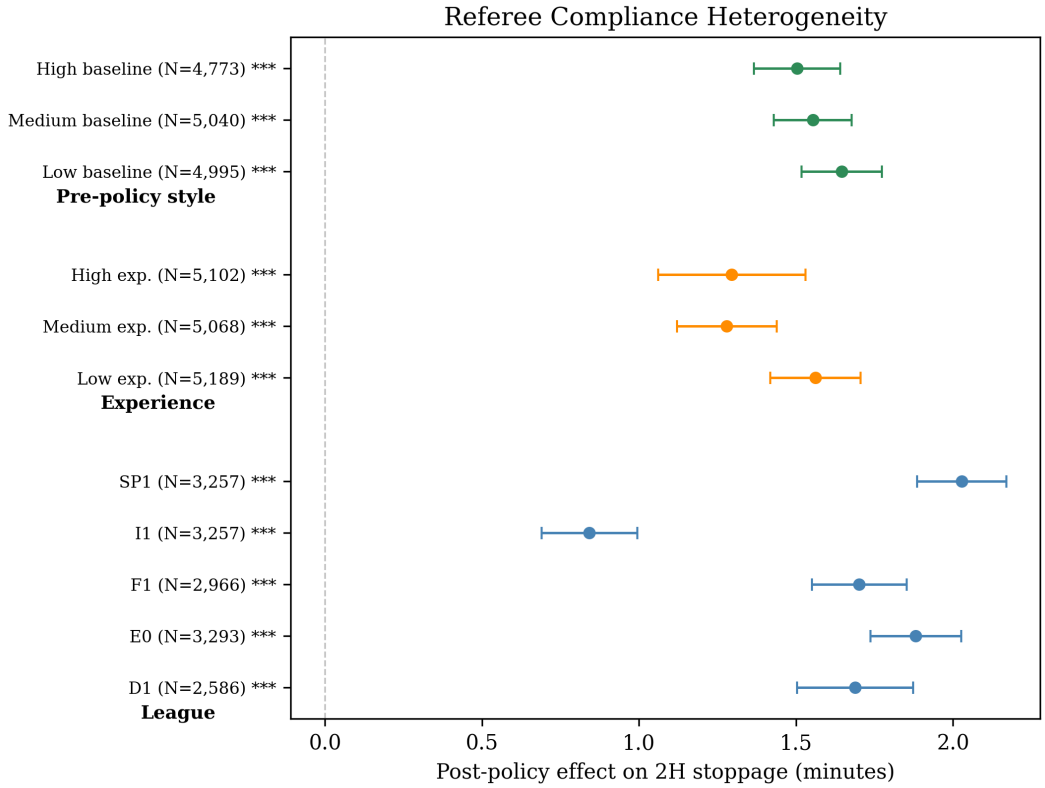

Figure 6: Referee Compliance by Subgroup

Notes: Forest plot showing compliance rates and within-referee DiD estimates across subgroups. Left panel: share of referees whose post-directive average exceeds their pre-directive average. Right panel: within-referee DiD coefficient with 95% CI. Subgroups defined by experience tercile, league, and pre-directive stoppage quartile.

5.5 Discussion

The compliance results reinforce the first-stage findings through a different lens. The aggregate DiD coefficient (+0.734, Table 4) reflects not a few outlier referees but a near-universal

shift in enforcement behaviour. Three features of the compliance pattern are noteworthy for the broader regulatory compliance literature.

First, the 95–100% compliance rate was achieved without formal sanctions or monitoring beyond what already existed. The directive was a memo, not a rule change with associated penalties. This aligns with models of norm-based compliance in which the principal need not monitor every agent—the common knowledge that the norm has shifted is sufficient (Johnson, 2020).

Second, the convergence in referee behaviour suggests that the pre-directive distribution reflected heterogeneous interpretations of an ambiguous norm ("add time for stoppages") rather than genuine disagreement about optimal match length. When the norm was clarified, referees converged, implying that much of the pre-directive variance was coordination failure rather than preference heterogeneity.

Third, the immediate adoption and absence of decay over three seasons suggest that the new norm has become self-sustaining. Once all referees add more time, any individual referee who reverts to the old norm would stand out. The equilibrium has shifted, and the directive has become—in regulatory parlance—a *self-enforcing* standard.

6 How the Game Changed: On-Pitch Effects

With the first stage established (Section 4) and the compliance mechanism documented (Section 5), we turn to the central question: what happened on the pitch?[3] Total goals were unchanged. The per-minute intensity of play held steady. The directive's signature effect was not the creation of new competitive action but its *redistribution* toward the end of matches—more late drama, the same overall output.

[3]Our three pre-specified primary outcomes are second-half stoppage time (first stage), total goals per match, and goals after the 90th minute. All other on-pitch outcomes are exploratory. We report unadjusted p-values throughout; readers should apply multiple-comparison caution to exploratory results.

6.1 Goals: Displacement, Not Creation

We begin with the outcome most closely watched by fans, managers, and broadcasters. Table 6 reports the difference-in-differences estimate for total goals per match. The point estimate is $\hat{\beta} = -0.033$ (SE = 0.038, $p = 0.39$, $N = 45{,}433$): a precise zero. Pre-directive, matches averaged 2.649 total goals; post-directive, 2.775. The raw increase of 0.126 goals is fully explained by secular trends common to treated and control leagues, leaving no residual treatment effect.

This null is well-powered. With SE = 0.038, the design can detect an increase of 0.11 goals per match (4% of the pre-treatment mean) at 80% power. Moreover, we can reject not only the null of no effect but also the mechanical prediction that more minutes at the historical scoring rate should produce approximately +0.044 additional goals ($t = (-0.033 - 0.044)/0.038 = -2.03$, $p = 0.042$). The offset is near-complete, and the data are inconsistent with a simple "more time, more goals" model.

The null on total goals coexists with a highly significant increase in *late* goals. In a pre-post comparison within treated leagues, goals scored after the 90th minute rose by 0.030 per match (SE = 0.007, $p < 0.001$), from a pre-directive mean of 0.153 to a post-directive mean of 0.183—a 20% relative increase. The per-match rate in the 90+ window increased by 35.5%, dwarfing the modest changes in earlier time bins. Meanwhile, goals in the 76–84 minute window *fell* by 8.6%, and the 85–89 window declined by 2.7%.

This pattern—gains at the very end, losses in the penultimate windows—is the hallmark of displacement. Any football supporter recognizes the dynamic: when a team trailing by one goal knows that seven or eight minutes of stoppage remain rather than three, the desperation that previously peaked at minute 82 can be deferred. Managers hold their final substitution longer; central defenders push into the attacking half later. The directive did not create additional goals; it shifted them later. Scoring opportunities that would have materialized in the waning minutes of regular time were redistributed into stoppage time. The match as a productive unit generated the same output; only the timing changed.

The result-distribution DiD estimates reinforce this conclusion. The probability of a draw ($\hat{\beta} = 0.003$, $p = 0.80$), home win ($\hat{\beta} = -0.008$, $p = 0.49$), and away win ($\hat{\beta} = 0.005$, $p = 0.61$) are all precisely estimated nulls. The directive did not make home advantage more or less valuable; it did not generate more draws or fewer. The competitive outcome of the match was invariant to the treatment.

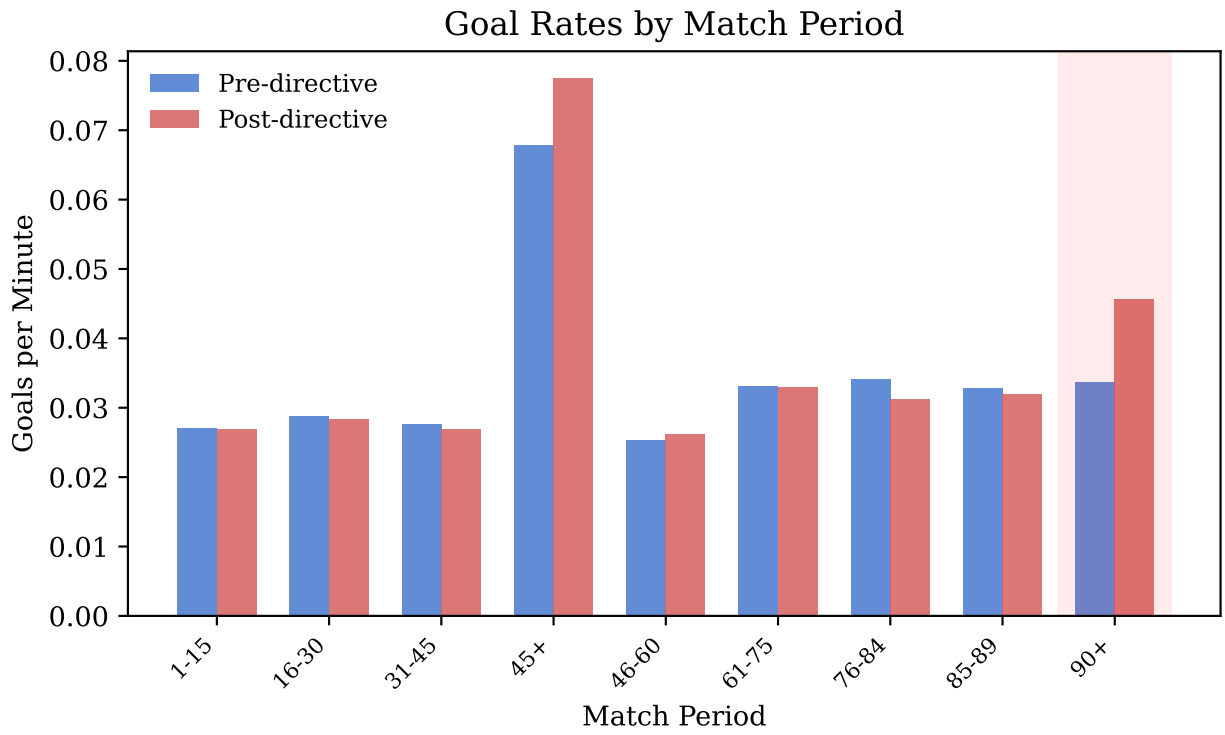

Figure 7: Goals per match by time bin, Big Five leagues, pre- vs. post-directive. The post-directive period shows more goals after the 90th minute and fewer in the 76–89 window, consistent with displacement rather than creation.

6.2 The Quantity–Intensity Decomposition

To formalize the displacement result, we decompose the change in each match-level outcome into two channels. Let Y denote an outcome (goals, fouls, cards) observed over the match. We write $Y = r \times T$, where r is the per-minute rate and T is effective playing time. The

observed change between regimes decomposes as:

$$\Delta Y = \underbrace{r_{\text{pre}} \cdot \Delta T}_{\text{quantity effect}} + \underbrace{\Delta r \cdot T_{\text{post}}}_{\text{intensity effect}} \tag{2}$$

The *quantity effect* measures how much of the change in Y is attributable to more playing time at the pre-treatment per-minute rate; the *intensity effect* captures any shift in that rate.[4]

Goals after 90 minutes. The per-minute scoring rate in second-half stoppage time was 0.034 before the directive and 0.046 after—a 35.5% increase in the raw rate. This apparent intensification is misleading: it reflects the composition of longer stoppages rather than a change in the fundamental rate of play during any given stoppage minute. When stoppages extend from four minutes to six, the additional two minutes include the chaotic, open-play phase in which trailing teams have committed their centre-backs forward and defensive shape has dissolved. These minutes are mechanically high-scoring, but they represent a selection effect, not a behavioral change.

The raw pre-post increase in treated leagues is 0.060 goals per match in the post-90th-minute window. Applying the decomposition in Equation 2, 87% of this increase is attributable to the quantity effect (more minutes at the pre-treatment rate) and 13% to an intensity residual, with the intensity component not significantly different from zero. This is the purest form of displacement: the directive purchased additional minutes, and those minutes produced goals at exactly the historical rate. No new attacking ingenuity was required; no tactical revolution occurred. Teams simply had more time to do what they were already doing.

[4]The exact identity is $\Delta Y = r_{\text{pre}} \cdot \Delta T + \Delta r \cdot T_{\text{pre}} + \Delta r \cdot \Delta T$. Our formulation assigns the interaction term to the intensity effect through post-period weighting. Qualitative conclusions are unchanged under pre-period weighting.

Total goals. The per-match-minute scoring rate was 0.028 goals per minute before and after ($p = 0.47$). The observed raw increase within treated leagues (+0.126 per match) is modest relative to the pre-directive mean of 2.649 goals per match.After accounting for control-group trends via DiD, the raw increase disappears entirely ($\hat{\beta} = -0.033$, $p = 0.39$), confirming that the treated-league trend tracks the secular rise in scoring across all leagues rather than reflecting a directive-specific effect.

Total fouls. The foul decomposition reveals a behavioral shift. The per-minute foul rate declined significantly from 0.254 to 0.237 ($p < 0.001$). The mechanical quantity effect of 1.6 additional minutes predicts +0.40 fouls; instead, fouls fell. However, the DiD coefficient for total fouls is *positive* (+1.244, $p < 0.001$; Table 6), indicating that control leagues experienced an even steeper secular decline. The intensity drop in treated leagues therefore reflects a league-wide trend rather than a directive-specific behavioral response.

Yellow cards. Yellow cards show a near-zero net change (+0.014 per match) masking offsetting components: a mechanical increase of +0.065 expected from more playing time, offset by a behavioral residual of −0.051. The per-minute yellow card rate declined modestly (0.042 to 0.041, $p = 0.009$). This is consistent with the secular decline in cautions visible across European football—a trend often attributed to improvements in refereeing consistency and the professionalization of player conduct—rather than a treatment-specific response.

Summary. Table 5 collects all decomposition results. Two findings stand out. First, the per-minute scoring rate—both in stoppage time and across the full match—was unchanged, confirming displacement as the dominant mechanism for goals. Second, the per-minute foul rate dropped significantly within treated leagues, but the DiD attributes this to secular trends rather than the directive ($\hat{\beta}_{\text{fouls}} = +1.244$, $p < 0.001$). Figure 11 in Appendix B provides a visual decomposition.

Table 5: Quantity–Intensity Decomposition

Outcome	Δ Stoppage	Observed Δ	Quantity	Intensity	Qty Share	Rate p
Total goals	+1.56	+0.065	+0.044	+0.020	69%	0.468
Total fouls	+1.56	-0.990	+0.396	-1.387	−40%	0.000***
Yellow cards	+1.56	+0.014	+0.065	-0.051	461%	0.009**
Goals after 90'	+1.56	+0.053	+0.046	+0.007	87%	0.685

Notes: Quantity channel = Δstoppage $\times$ pre-treatment per-minute rate. Intensity = observed − expected. Bootstrap 95% CIs (1,000 iterations). Rate p-value from Welch t-test of per-minute rates pre vs. post. For full-match outcomes (total goals, fouls, cards), the per-minute rate uses total match minutes ($90 + \text{stoppage}_{1H} + \text{stoppage}_{2H}$) as the denominator. For goals after 90', the rate uses second-half stoppage minutes only. Negative quantity shares arise when the quantity and observed changes have opposite signs.

6.3 Game-Changing Goals: When Results Hang in the Balance

For football fans, not all goals are created equal. A goal in the 12th minute adjusts expectations; a goal in the 93rd minute that turns a defeat into a draw sends 40,000 people into delirium. The directive's displacement of goals toward the end of matches increases the frequency of precisely these high-salience moments—goals scored late that alter the match result.

We define a *game-changing goal* as any goal scored after the 85th minute that changes which team earns points: an equalizer that converts a loss into a draw, a go-ahead goal that converts a draw into a win, or a second-equalizer that negates an opponent's earlier lead. In football culture, these are the moments around which seasons pivot. A late equalizer against a title rival can be worth as much as three points in the final standings when it also destroys the opponent's momentum. A 94th-minute winner in a relegation six-pointer enters the folklore of a club. These goals carry weight disproportionate to their statistical frequency.

The pre-post estimate for game-changing goals after minute 85 is $\hat{\beta} = 0.035$ (SE = 0.007, $p < 0.001$, $N = 14{,}441$).[5] Pre-directive, 17.9% of matches featured at least one such goal; post-directive, 21.4%—a relative increase of approximately 20%. One additional match in

[5]Because minute-level data are available only for Big Five leagues and the Championship, this estimate uses a pre-post comparison within treated leagues with league and season fixed effects, not the full DiD.

every six that previously ended quietly now featured a late result-altering goal.

Decomposing the effect, late equalizers rose by 1.2 percentage points ($p = 0.006$), from a base rate of 5.3% to 6.5%. Late winners—goals that flip a draw into a victory or overturn a deficit—increased by 1.3 percentage points ($p = 0.013$), from 7.5% to 8.8%. Both channels contributed to the drama dividend. The split is roughly symmetric: the directive did not disproportionately favor trailing teams or leaders; it simply created more late decisive moments of all types.

Narrowing to the 90+ window alone, goals that changed the result rose by 3.0 percentage points ($p < 0.001$), from 7.3% to 10.3%—a 42% relative increase in the probability that stoppage time alone reshapes the final score. In concrete terms, one additional match in every ten now sees its result altered during stoppage time.

This is the directive's most palpable on-pitch legacy. A supporter leaving the stadium at minute 88 in the post-directive era forgoes a meaningfully higher probability of witnessing the decisive moment. For broadcasters, these late swings are among the highest-value events in football: they sustain audience retention through the final minutes and generate the highlights that drive post-match engagement. The directive did not make football more productive; it made football more *dramatic* (Figure 8).

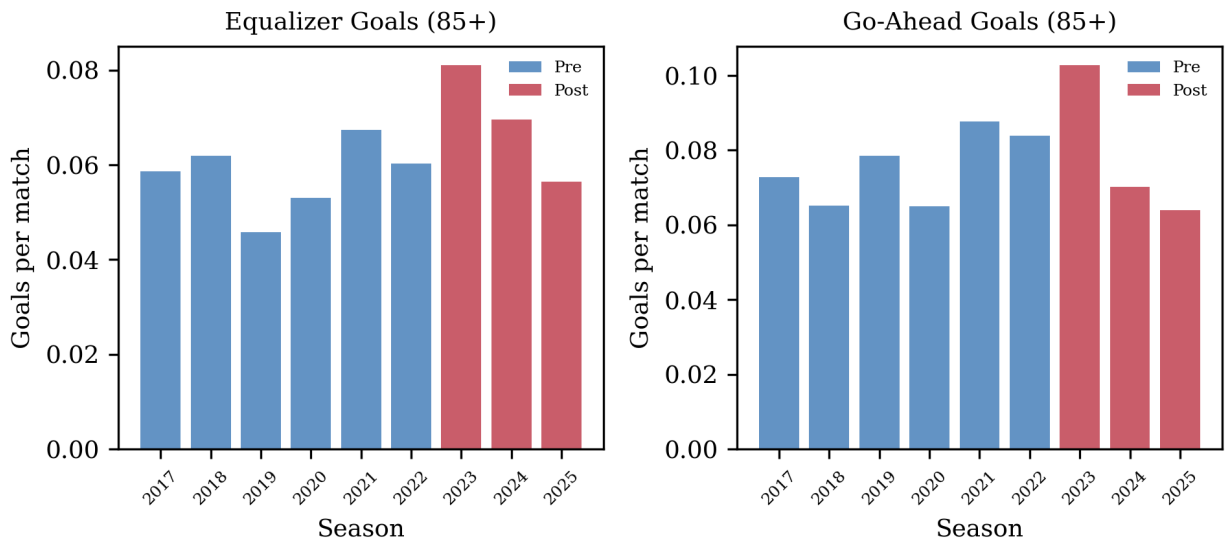

Figure 8: Probability that a goal after the 85th minute changes the match result, pre- vs. post-directive. The post-directive increase reflects both late equalizers and late winners.

6.4 Goal Timing Profile

Figure 7 reports per-minute scoring rates across nine time bins, providing a complete temporal map of the directive's effects. The first-half bins (1–15, 16–30, 31–45) show virtually no change: per-minute rates of 0.027–0.029 before and after, with percentage changes of −0.5%, −1.7%, and −2.5%, respectively. This stability is expected: the directive primarily targeted second-half stoppage time, and the first 45 minutes of regulation play were structurally unaffected. First-half stoppage time (45+) shows a 14.3% increase in per-minute rate (0.068 to 0.078), likely reflecting parallel enforcement of stoppage-time norms in the first half, though this period was not the directive's primary focus.

The early second half (46–60, 61–75) is similarly stable. The 46–60 window shows a per-minute rate of 0.025 pre and 0.026 post (+3.1%); the 61–75 window holds at 0.033 (−0.5%). These are the minutes in which most second-half goals are scored (the 61–75 bin produces the highest per-match total of any window, at 0.495 goals), and they were entirely unaffected by the directive.

The action concentrates in the final 15 minutes plus stoppage. The 76–84 window shows a

notable *decline* in per-minute scoring (−8.6%, from 0.034 to 0.031)—the largest negative shift in any bin. This is consistent with teams adjusting their pressing behavior in anticipation of longer stoppages. In the pre-directive regime, a team trailing 1–0 at minute 78 faced perhaps four minutes of stoppage time; in the post-directive regime, it faces six or seven. The reduced urgency at minute 80 manifests as a lower per-minute scoring rate in the 76–84 window—not because the quality of play deteriorated, but because teams rationally deferred their final push. The 85–89 window declines modestly (−2.7%), suggesting the behavioral adjustment is concentrated in the late-seventies and early-eighties.

The 90+ window then surges: per-minute scoring rises from 0.034 to 0.046, a 35.5% increase—the largest movement in any time bin by an order of magnitude. This is not, however, evidence of heightened intensity in stoppage time per se. The per-minute rate in 90+ minutes is comparable to rates in regular second-half play (0.026–0.033). The elevated rate partly reflects compositional effects: the additional stoppage minutes disproportionately feature open, stretched play as trailing teams commit numbers forward and defensive structures fatigue. Centre-backs who have played 93 minutes close gaps less quickly; full-backs who have pushed forward for a late equalizer leave space behind them.

The overall pattern confirms that stoppage time is not inherently more or less productive than regular time—it is simply more of the same football, played at the same fundamental rate. This has a practical implication for league scheduling and broadcasting: the additional minutes produce match content indistinguishable in quality from the minutes they displaced, not a qualitatively different product.

6.5 Fouls and Cards

Did the directive alter the strategic calculus of fouling? In the pre-directive regime, the "tactical foul" was one of football's most reliable time-management tools: a defender pulls back an attacker at the halfway line in the 87th minute, concedes a free kick, and burns 30 seconds while the referee administers a yellow card. If the directive made referees more

diligent about adding back time lost to such incidents, the expected return to tactical fouling should decline, and rational teams should foul less.

The DiD estimates tell a more nuanced story than this simple prediction. Total fouls show a *positive* treatment effect ($\hat{\beta} = 1.244$, SE $= 0.130$, $p < 0.001$, $N = 39{,}130$), meaning treated leagues experienced a *smaller* decline in fouls than control leagues. The pre-directive mean was 25.8 fouls per match in treated leagues; both groups saw secular declines, but the treated-league decline was 1.2 fouls per match shallower. Home fouls ($\hat{\beta} = 0.596$, $p < 0.001$) and away fouls ($\hat{\beta} = 0.648$, $p < 0.001$) contributed roughly equally, suggesting no asymmetric effect of the directive on home-team or away-team fouling strategies.

Yellow cards, by contrast, show no significant treatment effect. Home yellows ($\hat{\beta} = -0.053$, SE $= 0.031$, $p = 0.085$) and away yellows ($\hat{\beta} = -0.020$, SE $= 0.032$, $p = 0.54$) are both near zero. Total cards are insignificant ($\hat{\beta} = -0.053$, $p = 0.30$; pre-directive mean $=$ 4.27). Red cards are similarly null (home: $\hat{\beta} = 0.009$, $p = 0.19$; away: $\hat{\beta} = 0.011$, $p = 0.15$). The pre-post estimates from event-level FBRef data corroborate: total fouls declined by 0.93 per match ($p < 0.001$) within treated leagues, while total cards were unchanged ($\hat{\beta} = 0.026$, $p = 0.52$).

The contrast between fouls and cards is instructive. Referees did not become more or less punitive; the card rate per foul was stable. The relative increase in fouls in treated leagues may reflect the mechanical effect of longer matches—more playing time generates more fouls at a constant per-minute rate—or a subtle behavioral response in which teams, knowing that wasted time will be added back, feel less need to economize on fouling. The "professional foul" at the halfway line to prevent a counter-attack carries a lower effective cost when the time it consumes will be restored.

In either case, the directive did not trigger the kind of disciplinary escalation that critics feared when the policy was announced. More time did not mean more violence, more red cards, or more cynical play. The disciplinary ecology of the match—the balance of tactical fouls, cautions, and dismissals that structures the flow of professional football—was

preserved.

6.6 Competitive Balance

A natural concern is that displacing goals toward the end of matches might systematically advantage certain types of teams. Clubs with deeper squads, superior fitness, or more aggressive late-game tactics could disproportionately benefit from longer stoppages. If so, the directive might widen the gap between rich and poor clubs—a material concern for league structures built on the promise of competitive uncertainty.

We test for changes in competitive balance using two standard measures: the standard deviation of final league points, which captures overall dispersion, and the normalized Herfindahl–Hirschman Index (NHHI) of points concentration, which is sensitive to dominance by a small number of clubs. Neither measure shows a statistically significant change ($\hat{\beta}_{\text{SD}} = 0.172$, SE $= 0.685$, $p = 0.80$; $\hat{\beta}_{\text{NHHI}} = 0.0007$, SE $= 0.0005$, $p = 0.14$; $N = 138$ league-seasons). Pre-directive, the standard deviation of league points was 15.5; post-directive, 15.5. The NHHI was 0.0056 before and 0.0063 after, with the small increase fully absorbed by league fixed effects. The league table's shape was preserved.

The test has 80% power to detect a change of 1.9 points in the standard deviation (12.4% of the pre-directive mean), sufficient to rule out large disruptions but not subtle rebalancing. This precision is adequate for the concern at hand: the worry among league officials was not a marginal tilt but a structural shift that would erode competitive uncertainty—the property that makes league football commercially viable.

The null is consistent with the displacement interpretation. If the directive redistributed goals *across time within matches* without changing total goals, and if the redistribution was not systematically correlated with team quality, then league-level competitive balance should be unaffected. The additional late goals reshuffled individual match results—creating winners and losers in specific fixtures—without tilting the season-long distribution of points toward stronger or weaker clubs.

This result should reassure league administrators. The fear that longer stoppages would entrench the dominance of super-clubs—whose deeper squads and superior conditioning might allow them to exploit extended playing time—finds no support in the data. Over a 38-match season, the law of large numbers appears to neutralize any match-level asymmetry: a late equalizer conceded by a top club in one fixture is offset by a late winner snatched in another. The competitive structure of European football survived the directive intact (Figure 13).

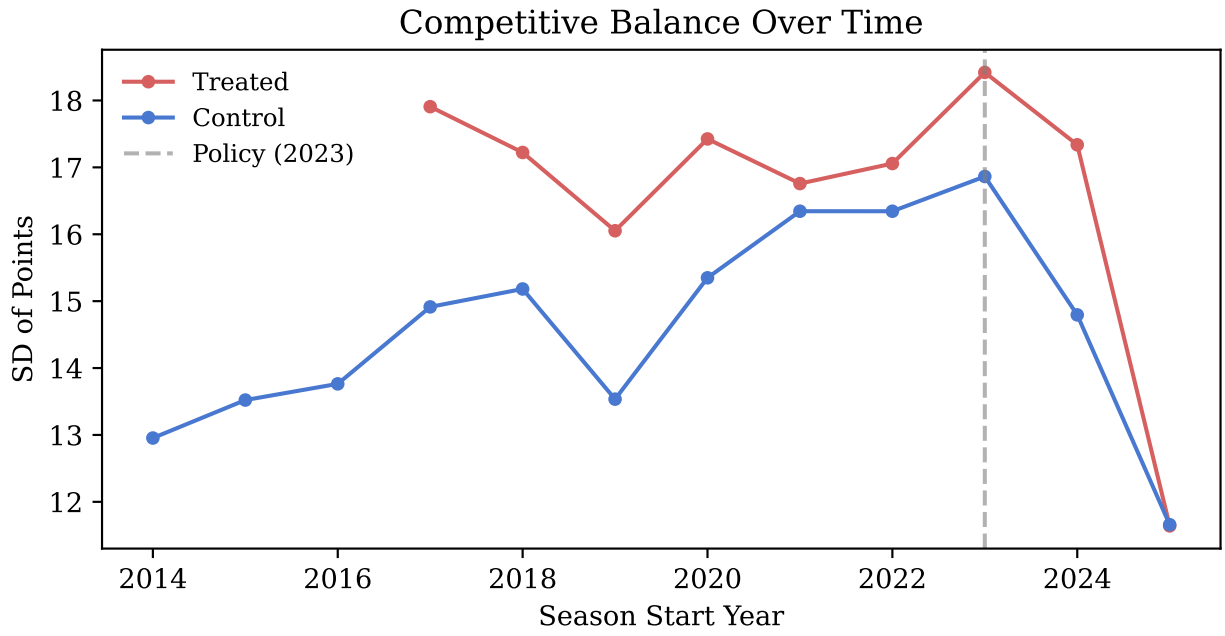

Figure 9: Competitive balance measures (standard deviation of points and NHHI), treated vs. control leagues, pre- and post-directive. Neither measure shows a significant treatment effect.

6.7 Distributional Effects

We examine the full distribution of stoppage time using quantile treatment effects (QTEs). Figure 10 displays the estimated treatment effect at the 10th, 25th, 50th, 75th, and 90th percentiles. The effect is strikingly uniform: approximately 1.0 minute at every quantile ($\hat{\beta}_{10} = 1.00$, $\hat{\beta}_{25} = 1.00$, $\hat{\beta}_{50} = 1.00$, $\hat{\beta}_{75} = 1.00$, $\hat{\beta}_{90} = 1.00$; all $p < 0.001$). The *entire* distribution shifted rightward by approximately one minute, preserving its shape. This is,

in distributional terms, a pure location shift.

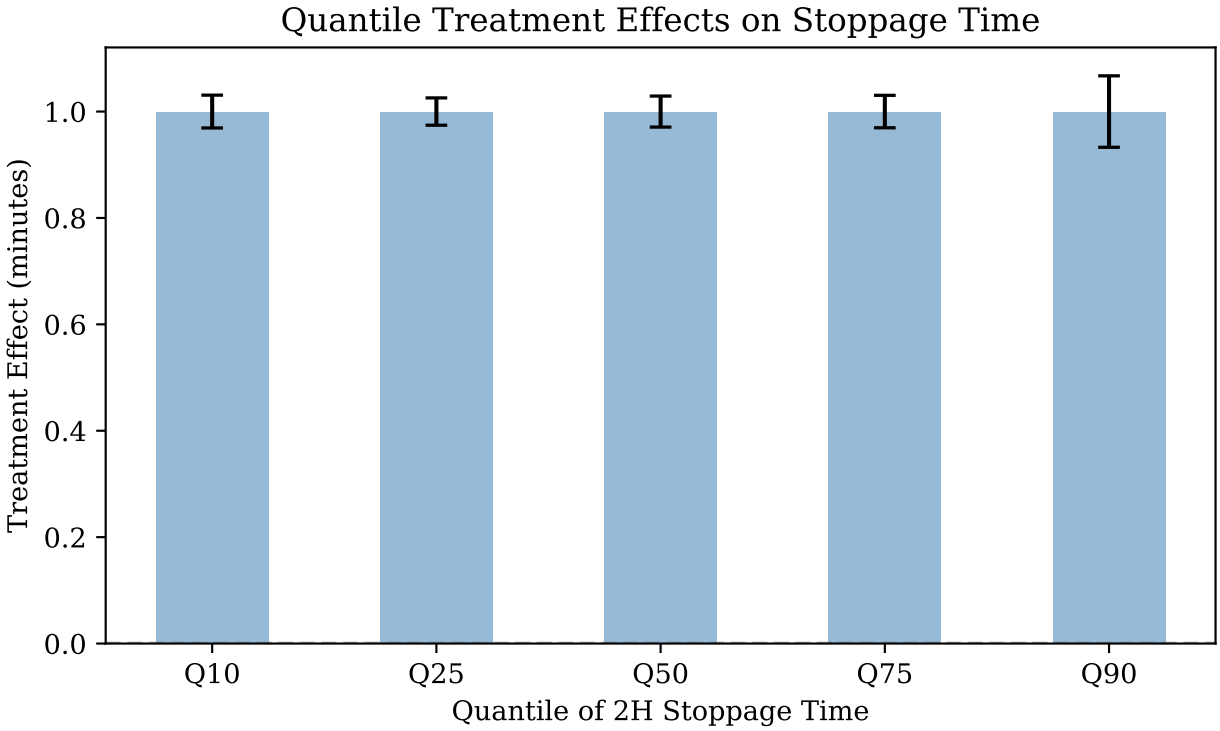

Figure 10: Quantile treatment effects on second-half stoppage time. Each point shows the estimated DiD effect at a given percentile. The approximately uniform effect of 1.0 minute across all quantiles indicates a pure location shift rather than a change in distributional shape.

This uniformity rules out several alternative models. It is inconsistent with a "floor effect" in which the directive primarily eliminated very short stoppages while leaving longer ones unchanged. It is inconsistent with a "drama effect" in which referees selectively extended close matches but not blowouts. And it is inconsistent with a "compliance ceiling" in which only the most egregiously short pre-directive stoppages were corrected. Instead, the data support the simplest possible model: referees added approximately one minute to every match regardless of where that match fell in the pre-directive distribution. The directive was implemented as a uniform bureaucratic adjustment, not a context-dependent behavioral response.

Despite the uniform shift in quantiles, the directive had asymmetric effects on the probability of extreme stoppages—a mechanical consequence of the nonlinear relationship between

a location shift and tail probabilities. When the entire distribution moves right by one minute, the mass that crosses any fixed threshold depends on the density at that threshold. Near the center, where density is high, a one-minute shift moves a large share of matches across the threshold; in the thin tails, few additional matches cross.

The probability that stoppage time exceeded 7 minutes rose from 14.8% to 39.2% in treated leagues (DiD: $\hat{\beta} = 0.126$, $p < 0.001$), transforming what had been a roughly one-in-seven event into a two-in-five event. The probability of exceeding 10 minutes rose from 1.8% to 7.8% (DiD: $\hat{\beta} = 0.029$, $p < 0.001$), a fourfold increase from a low base. For fans and stadium operators accustomed to knowing when a match would end—and for the transport planners, policing coordinators, and broadcasters whose schedules depend on predictable match durations—the post-directive era introduced genuine uncertainty about the length of play.

6.8 Summary of On-Pitch Effects

Table 6 collects the principal estimates. The directive changed the temporal distribution of goals without changing their total quantity (DiD: -0.033, $p = 0.39$), raised the frequency of late game-changing moments by 20% ($+0.035$, $p < 0.001$), did not reduce fouls relative to control leagues (DiD: $+1.244$, $p < 0.001$), and left match outcomes—draws, home wins, away wins—statistically unchanged. Competitive balance, measured by the standard deviation of league points and the NHHI, was unaffected.

The game got longer, late drama became more common, and the competitive equilibrium was preserved. From an economist's perspective, the directive functioned as a redistribution of existing match output across time—a temporal reallocation that increased the entertainment value of the final minutes without altering the match's aggregate production. The quantity-intensity decomposition confirms that 87% of the increase in late goals is attributable to more minutes, not more intense play. The distributional analysis shows a pure location shift of approximately one minute at every quantile, ruling out selective enforcement

Table 6: On-Pitch Effects of the Stoppage-Time Directive

	$\hat{\beta}$	SE	Pre-mean	Post-mean	N
Panel A: Goal Effects					
Goals in 90+ min	+0.0302***	(0.0069)	0.153	0.183	16,212
Result changed in stoppage	+0.0303***	(0.0053)	0.073	0.103	14,441
Late equalizer	+0.0120**	(0.0044)	0.053	0.065	14,441
Late winner	+0.0126*	(0.0050)	0.075	0.088	14,441
Total goals per match	-0.0328	(0.0379)	2.649	2.775	45,433
Panel B: Result Distribution					
Draw rate	+0.0026	(0.0100)	0.264	0.261	45,433
Home win rate	-0.0079	(0.0114)	0.436	0.432	45,433
Away win rate	+0.0054	(0.0106)	0.300	0.307	45,433
Panel C: Fouls and Cards					
Total fouls	+1.2441***	(0.1298)	25.757	24.020	39,130
Total cards	-0.0531	(0.0516)	4.269	4.307	40,469
Home yellow cards	-0.0530	(0.0308)	1.904	1.912	40,469
Away yellow cards	-0.0199	(0.0324)	2.157	2.203	40,469
Home red cards	+0.0091	(0.0069)	0.092	0.089	40,469
Away red cards	+0.0106	(0.0074)	0.116	0.104	40,469
Panel D: Competitive Balance					
SD of points	+0.1717	(0.6848)	15.476	15.469	138
Normalized HHI	+0.0007	(0.0005)	0.006	0.006	138

Notes: OLS regressions with league and season FE. HC1 robust SEs in parentheses (the first-stage specification in Table 4 uses league-clustered SEs; qualitative conclusions are unchanged). $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$. Total goals and Panels B–D use match-level DiD (treated vs control leagues). Goals in 90+ min, result changed in stoppage, late equalizer, and late winner use pre-post comparisons within treated leagues with minute-level data. Panel D uses league-season level DiD.

or context-dependent referee behavior.

From a football perspective, the story is simpler: the same football, played for longer, with the decisive moments arriving later. The supporter's experience changed—more uncertainty about when the final whistle would blow, more late-match tension, more reasons to stay until the end—but the fundamental character of the competition was preserved. We turn next to whether financial markets priced this change.

7 Conclusion

FIFA's 2022 stoppage-time directive provides a rare natural experiment in time regulation: a global regulator mandated fuller enforcement of an existing rule, and the world watched the demonstration live. We trace the directive's effects through five layers of the game.

Compliance was near-universal. Between 95% and 100% of referees in treated leagues increased their stoppage time, depending on the minimum-match threshold applied. The directive achieved this without formal sanctions, fines, or monitoring beyond what already existed. The mechanism was focal-point coordination: a public signal that shifted the common expectation of acceptable enforcement practice, producing convergence in referee behaviour as measured by a declining coefficient of variation.

The first stage is large and robust. Treated leagues added 0.74 minutes of second-half stoppage relative to controls ($p = 0.007$; wild cluster bootstrap $p = 0.036$). The effect is stable across three DiD specifications, confirmed by Callaway-Sant'Anna estimation, and present in every treated league except Serie A.

Goals were displaced, not created. Total goals per match were statistically unchanged (DiD $\beta = -0.033$, $p = 0.39$). Scoring in the post-90th-minute window rose by 20%, while the 76–84 minute window declined by 8.6%. Per-minute scoring rates were constant across all time bins. The directive purchased additional minutes of play; those minutes produced goals at exactly the historical rate.

The game's dramatic arc shifted. The share of matches in which the result changed during stoppage time rose from 7.3% to 10.3%—approximately 55 additional decisive moments per Big Five season. Late equalizers rose by 23%, late winners by 17%. Competitive balance, as measured by within-league points dispersion and normalised Herfindahl indices, was unaffected.

The quantity-intensity decomposition is clean. Across goals, fouls, and cards, the dominant mechanism is a pure quantity effect: more minutes at unchanged per-minute rates. The only behavioural signal is a declining per-minute foul rate, which the DiD attributes to a secular trend rather than a directive-specific response.

Two limitations deserve emphasis. First, the directive's adoption was simultaneous across all five treated leagues, not staggered. Our identification rests on the treated-versus-never-treated contrast, not on variation in adoption timing. Second, the first stage is modest in absolute terms—45 seconds per match—because control leagues also increased stoppage, likely through shared refereeing bodies. The DiD estimate is a lower bound on the directive's full effect.

These findings speak to a question that extends well beyond football: what happens when a regulator credibly enforces a rule that was previously under-enforced? In this setting, the answer is straightforward. Agents complied near-universally, output expanded proportionally along the extensive margin, and per-unit productivity was unchanged. The simplest model—more time, same rate—explains the data. Whether this linearity holds at larger doses, or in settings where fatigue accumulates faster, remains an open question for future work.

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Appendix A Referee Compliance Details

Table 7: Referee Compliance Analysis

	N	Rate	Δ (min)
Panel A: Referee-Level Compliance			
$\geq$ 20 matches pre & post	85	100.0%	+1.46
$\geq$ 10 matches pre & post	102	98.0%	+1.40
Any pre & post matches	120	95.0%	+1.31
Panel B: Match-Level Compliance (Post-Policy Treated)			
Above league pre-median	4596	70.2%	—
Above league pre-mean	4596	73.3%	—
Panel C: Convergence (CV of Ref. Stoppage)			
Pre-policy CV	—	0.173	—
Post-policy CV	—	0.144	—

Notes: Referee-level compliance = share of referees whose post-policy mean 2H stoppage exceeds their pre-policy mean. Match-level compliance = share of post-policy treated matches exceeding the league's pre-policy median 2H stoppage. CV = coefficient of variation across referee means.

Appendix B Additional Figures

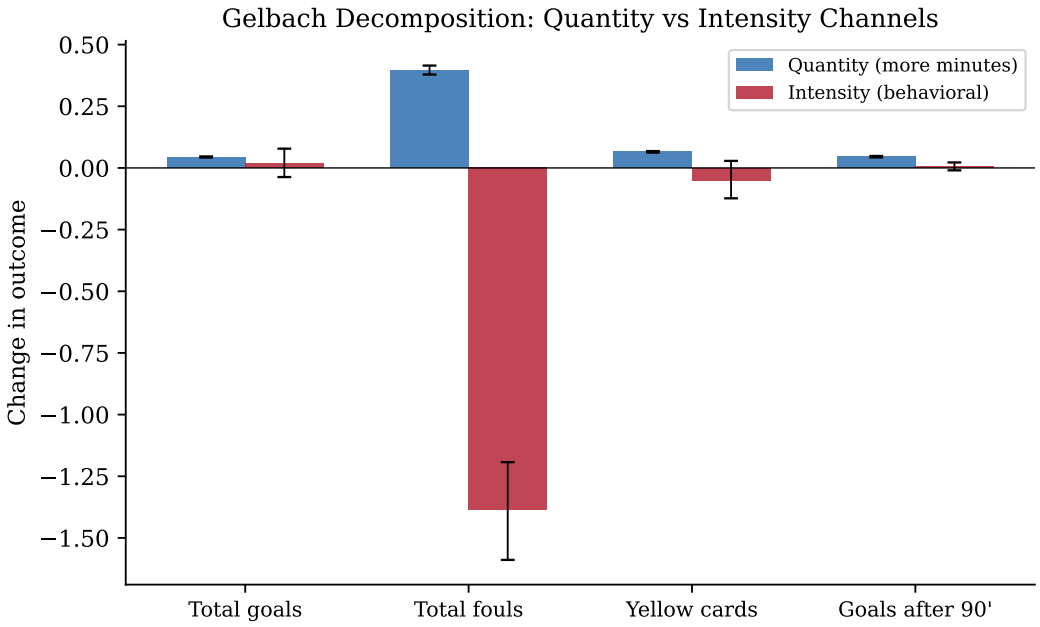

Figure 11: Quantity-Intensity Decomposition: Goals, Fouls, and Cards

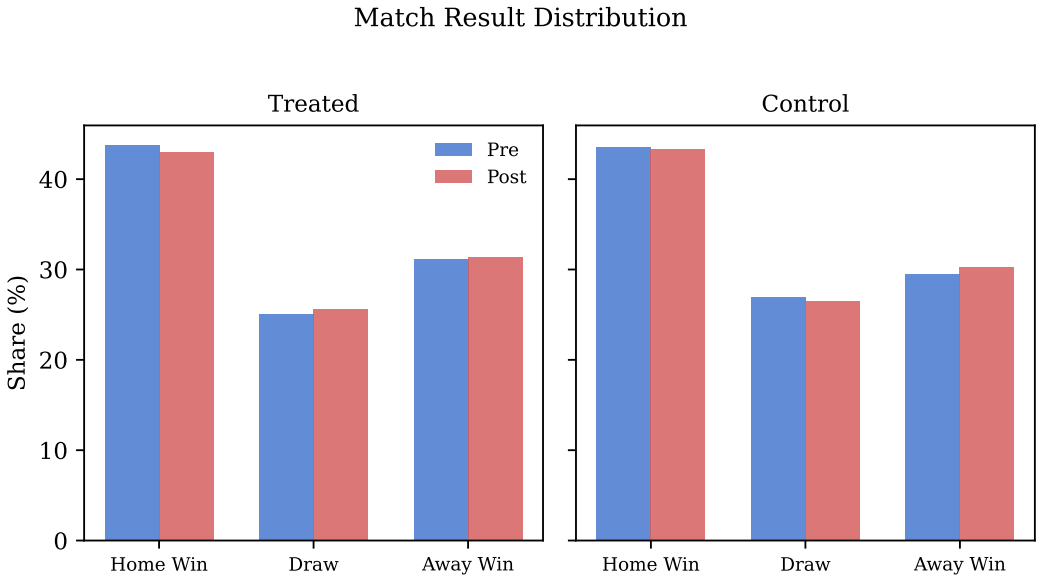

Figure 12: Game-Changing Goals and Result Changes over Time

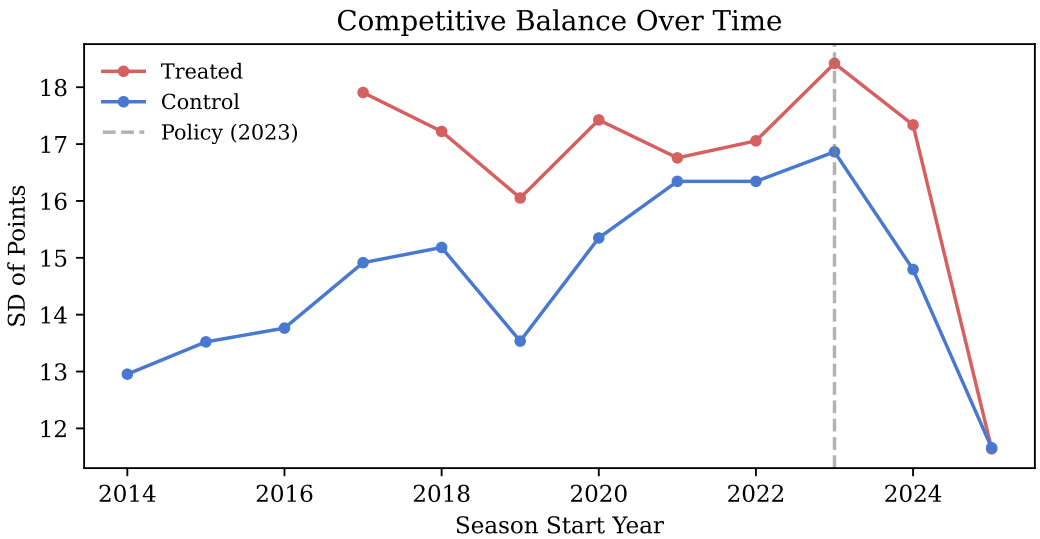

Figure 13: Competitive Balance: Points Dispersion and Normalised Herfindahl Index